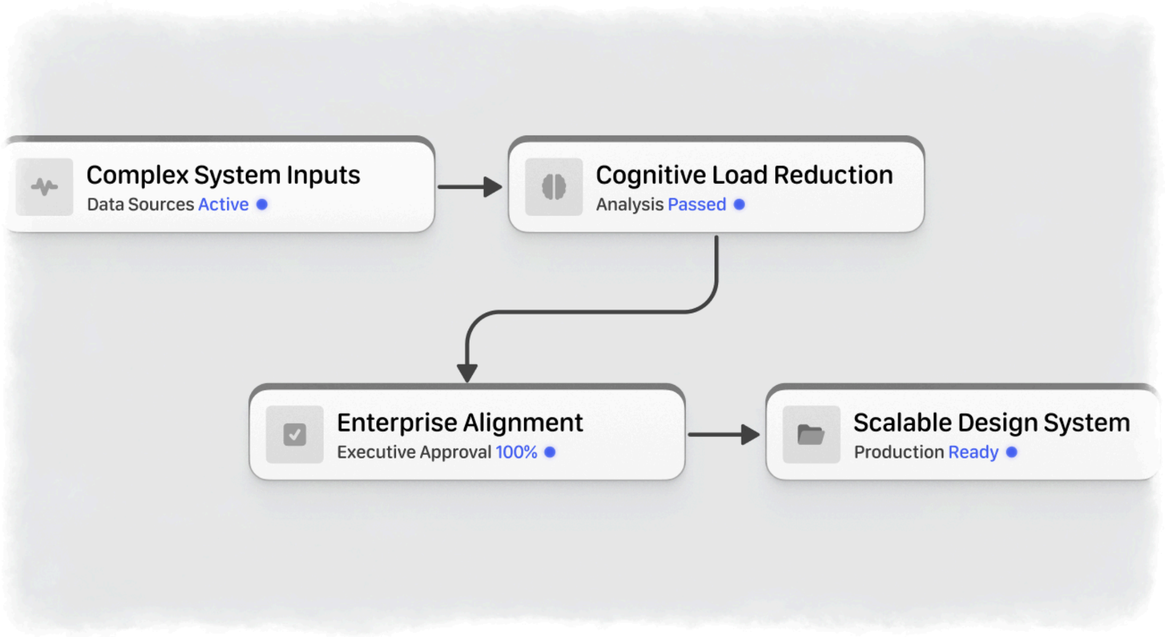
Floria Léger

Cognitive UX Designer

Bridging cognitive engineering and visual craft to simplify mission-critical software, enterprise design systems, and complex human-machine workflows.

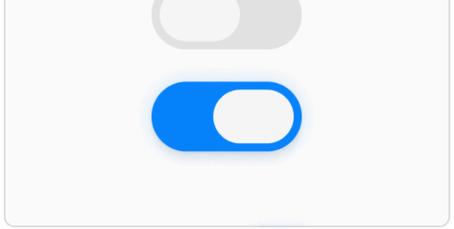


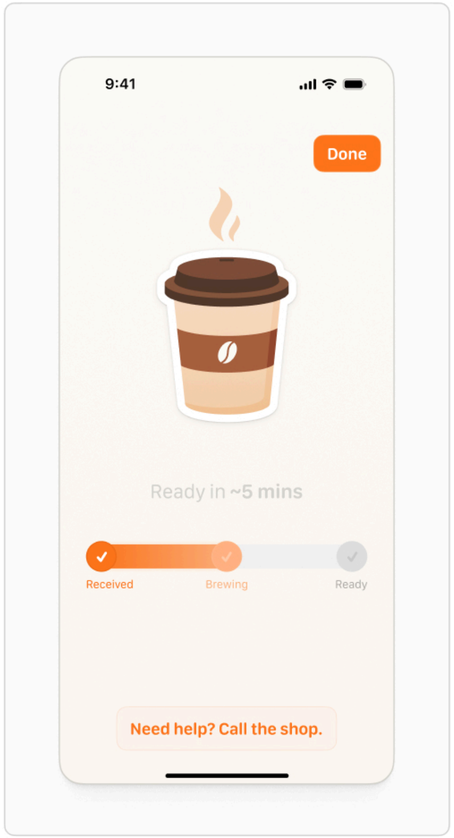
All visuals © Floria Léger

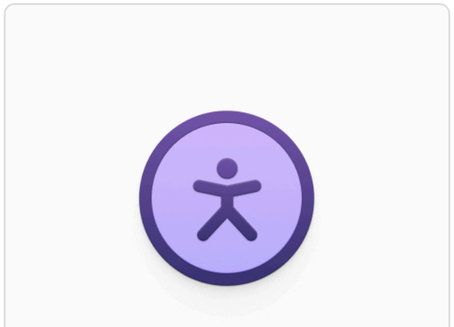
01 About me

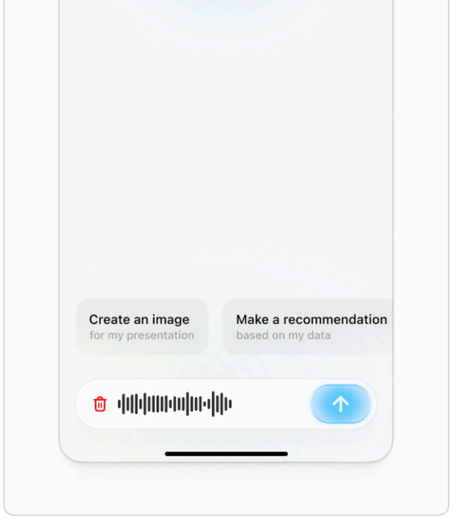
I am a Master's graduate in UX Design, Cognitive Science, and Computer Science from the National Superior School of Cognitics (ENSC Bordeaux). I combine cognitive ergonomics, empirical user research, and technical software principles to craft scalable digital products.

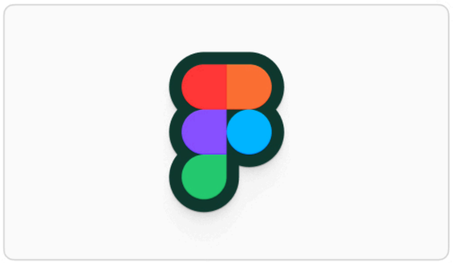
Backed by over three years of dedicated product design practice and 18 years of visual art discipline, I design systems that align with human mental models, streamline developer handoffs, and scale cleanly across engineering teams.

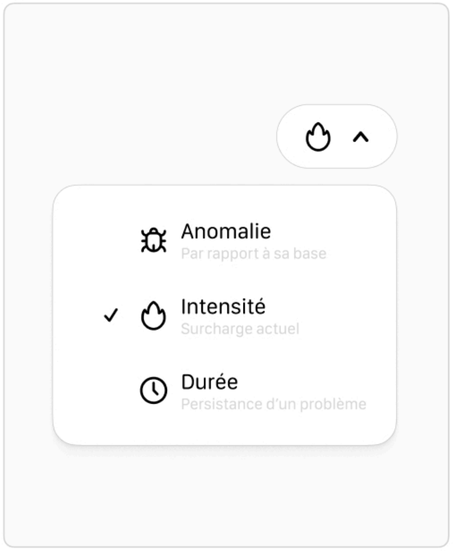


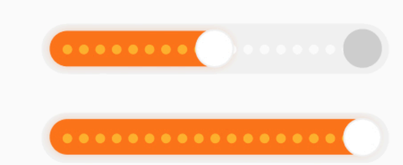


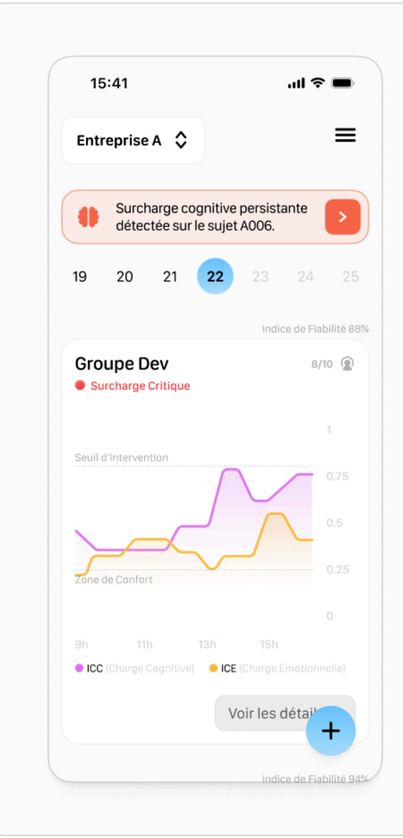


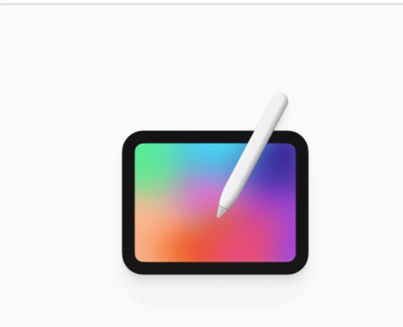












Cognitive Load Minimization

Complex software should not require a user manual. I structure user interface architectures around human working memory limits to eliminate operational friction and decision fatigue.

Function-Driven Visual Hierarchy

Aesthetics must always drive functional clarity. I leverage spatial hierarchy, dynamic typography, and data visualization to translate dense telemetry into immediate, confident action.

Systemic Engineering Rigor

Interface consistency establishes operational trust. I architect modular Figma design systems with tokenized Auto Layout components that accelerate development velocity and maintain design integrity.

03 Experiences & Core Competencies

Cognitive Research

Methods: Contextual Inquiry, Qual/Quant Research, Usability Protocols
Analysis: Mental Model Mapping, Cognitive Load, AttrakDiff
Eval: Heuristic Audits (10-Criteria), Information Architecture

Systems Architecture

UI Systems: Enterprise Design Systems, Tokens (Auto Layout 5.0)
Specs: Hi-Fi Prototyping, Micro-Interactions, WCAG Standards
Spatial UX: 2D Overlays for 3D/CAD Viewports

Strategy

Alignment: RACI Frameworks, Operational Strategy
Decision: Impact/Effort Matrices, Product Benchmarking

Software Mastery

Design: Figma, Procreate, Motion Graphics, Vector
Dev Stack: HTML5, CSS3, JavaScript, React, Node, Git

UX Designer Intern Ubisoft 2026, Paris

Driven under the guidance of my Design Manager, I led the UX research and operational architecture for global Creator Visibility across six corporate departments, transforming unexploited partner content into scalable social marketing assets while securing 100% executive alignment.

Lead UX Designer Stealth Startup 2025 - 2026, Bordeaux

Co-led product strategy by executing competitive market benchmarks and user research, while engineering an enterprise-grade Figma design system with components and design tokens to accelerate front-end development velocity.

UX Consultant Intern Airbus & IMS Lab 2025, Bordeaux

Executed a corrective user-centered overhaul of the Virtual Maintenance Trainer interface by evaluating 75 total participant responses across 3 questionnaires (including a self-created UMUX survey), 7 qualitative instructor interviews, and CAD-compliant 2D overlay specifications.

UX Instructor Intern Bordeaux INP 2024, Bordeaux

Designed and facilitated interactive UX/UI workshops for undergraduate students, teaching Auto Layout component architectures, wireframing workflows, and cognitive design principles.

01 **Creator Visibility** Enterprise Workflow Architecture — Ubisoft

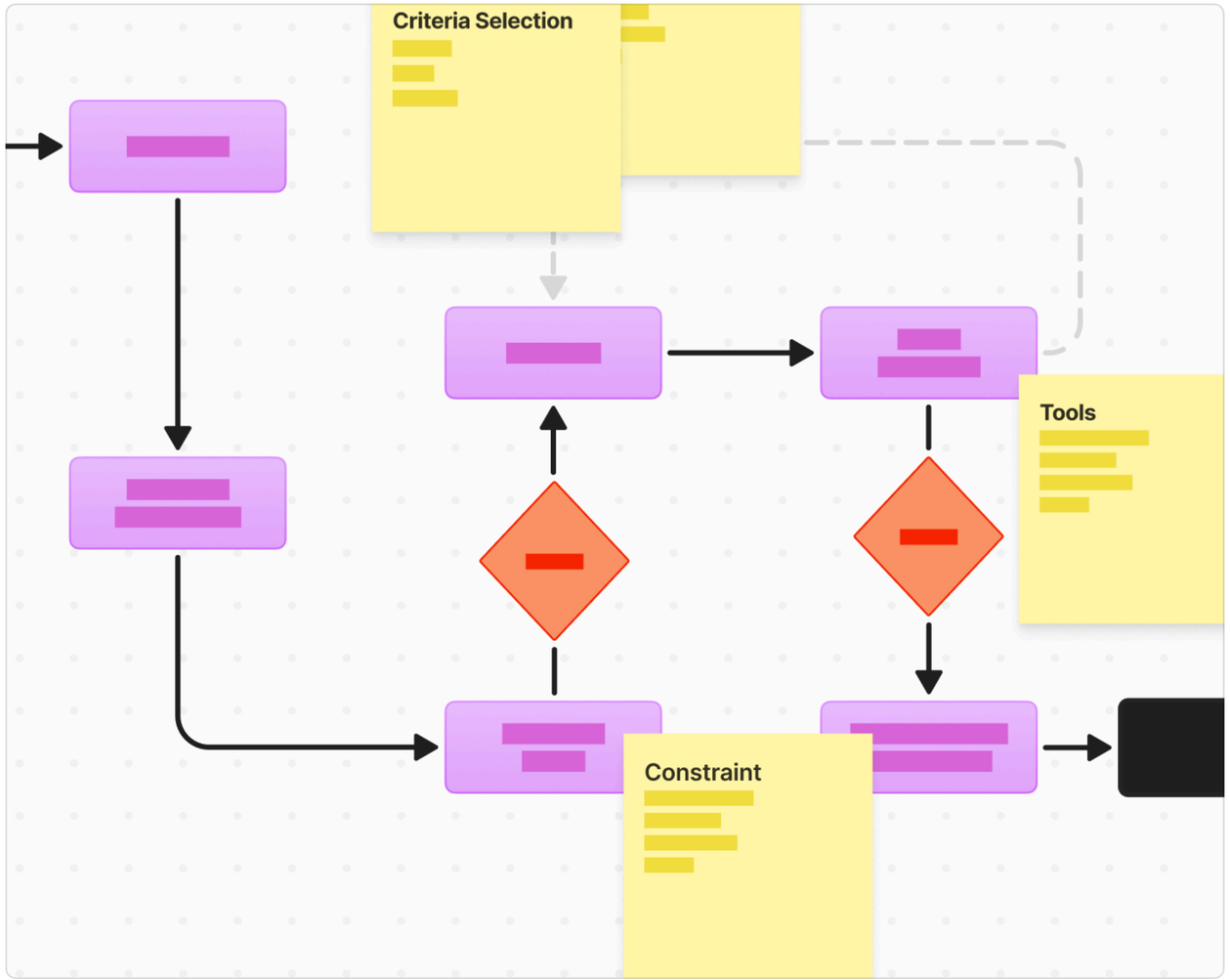
02 **VMT** Optimizing Cognitive Affordance in Industrial CAD Training — Airbus & IMS Lab

03 **EmCo** Decoupling Emotional State from Cognitive Workload — Academic

04 **Multi-Product UCD Sprint** Rapid Systems Architecture & Field Testing — Academic

05 Case Study 1: Creator Visibility Enterprise Workflow Architecture

Ubisoft (Online Services), Feb 2026 - July 2026, Paris



Presented under non-disclosure constraints. Proprietary internal tools and sensitive business metrics have been anonymized or generalized without altering core UX methodology.

Role: UX Designer & Researcher Intern

Domain: Cross-Departmental Operational Strategy, Creator Ecosystems & Governance

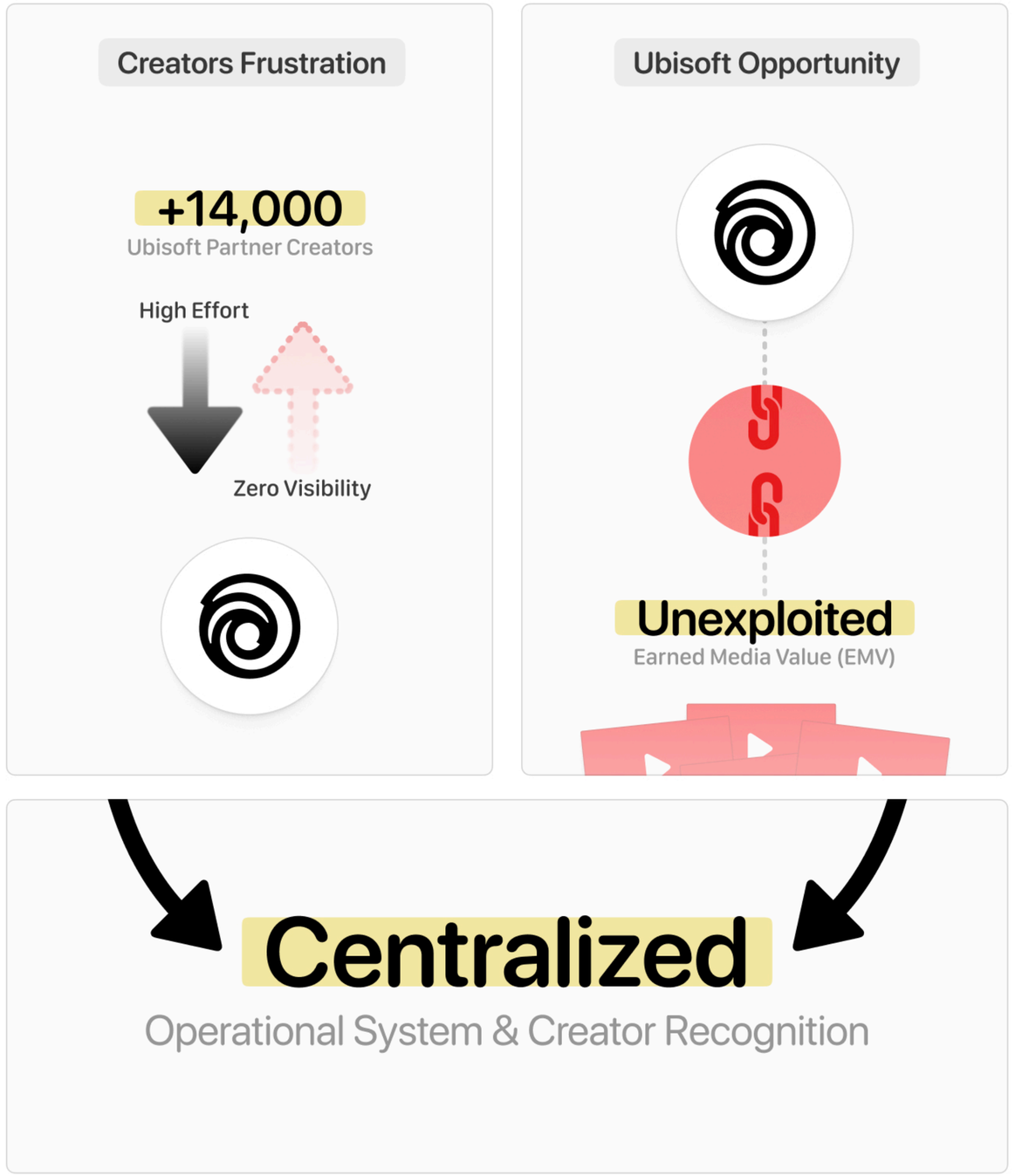
Human Friction

Within the Ubisoft Creator Program, over 14,000 partner creators felt invisible. Producing high-effort content with zero brand feedback directly eroded creator retention and community trust.

Business Opportunity

Massive volumes of partner content went unexploited. Ubisoft left significant potential Earned Media Value (EMV) on the table because internal teams lacked a structured approach.

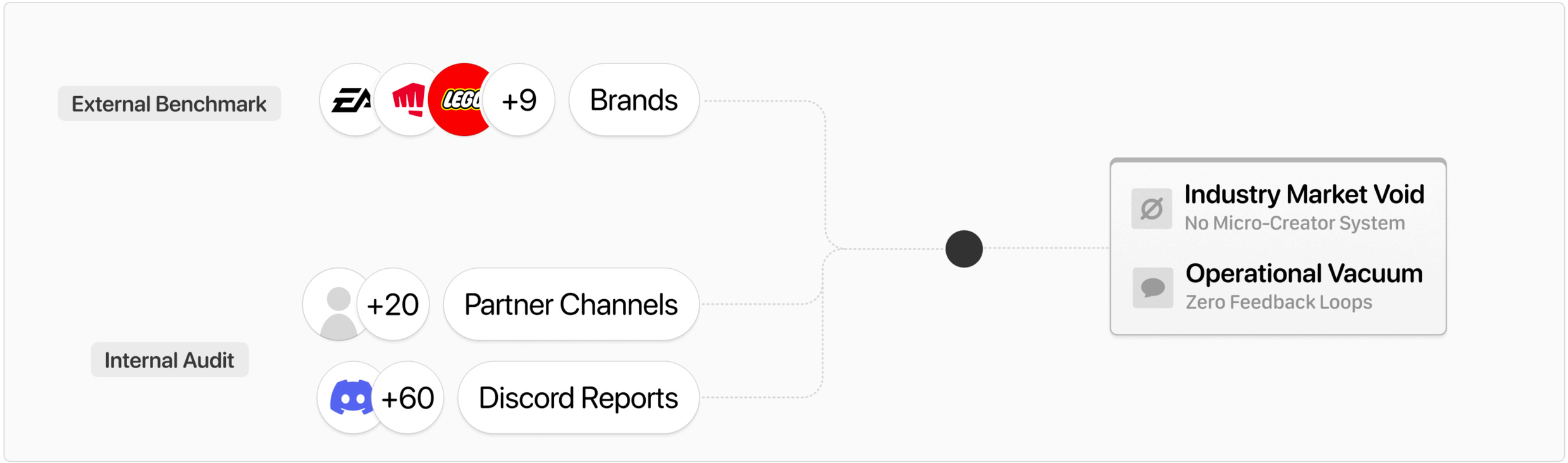
Working as the primary UX designer under the guidance of my Design Manager, my goal was to define an operational strategy that bridged this dual gap within strict enterprise constraints.



Discovery

To validate creator friction, I executed a two-pronged discovery phase. First, I conducted an external public benchmark across 12 entertainment brands (including EA, Riot, and LEGO). This revealed that while macro-influencers received dedicated PR spotlighting, no competitor offered a visible, structured recognition system for micro and nano-creators.

Concurrently, I audited 20+ partner channels and analyzed 60+ Discord community reports. This research confirmed that creator dissatisfaction stemmed from an operational void: creators produced content in a complete vacuum, lacking feedback loops or clear channels for internal discovery.



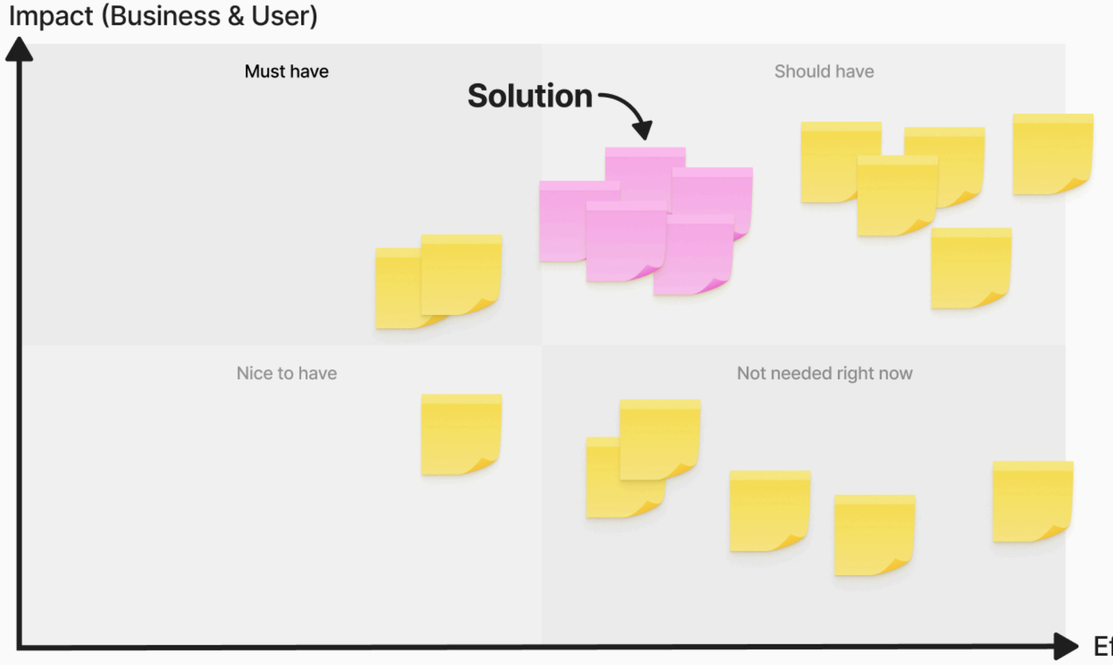
Cross-Functional Co-Design

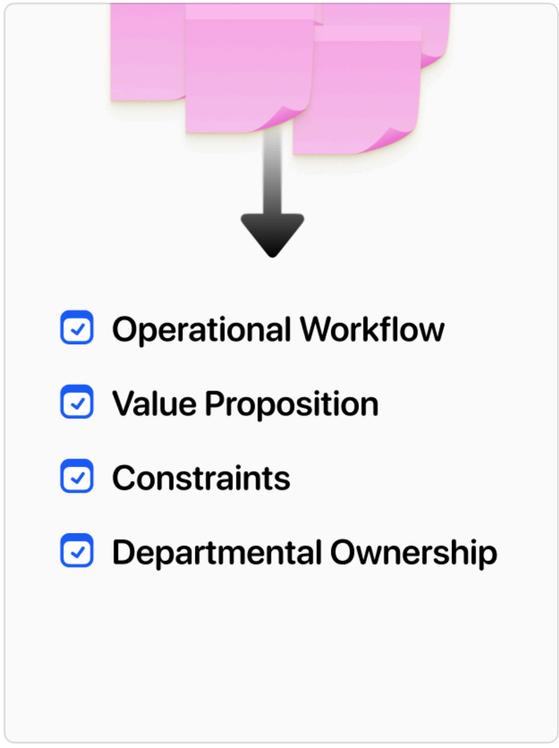
Because creator invisibility was rooted in backend operational confusion, I focused ideation on internal stakeholders rather than external creators who could not resolve internal handoff mechanics.

Strategic Frameworks

I structured and facilitated a 2-hour 20-minute workshop, inviting leads specifically for their daily proximity to partner creators and domain expertise.

Using an Impact/Effort matrix, we evaluated 24 concepts and converged on official channel reposting as our primary mechanism. This collaborative session also served to identify which key departments needed to be formally integrated into the eventual pipeline.





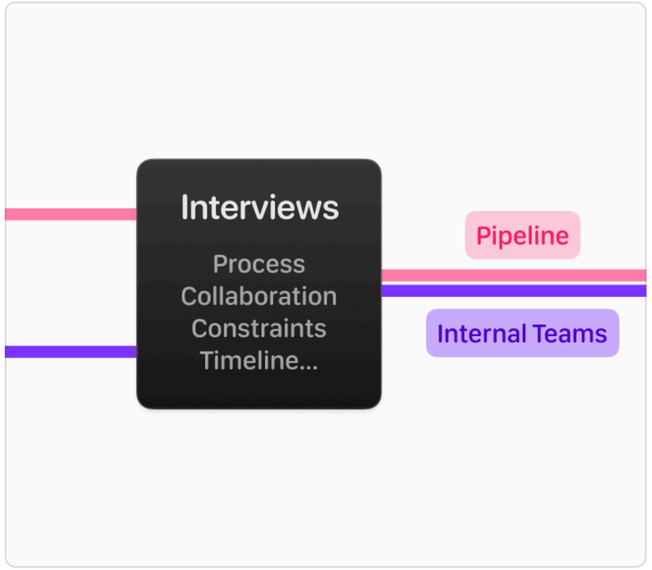
Workflow Refinement

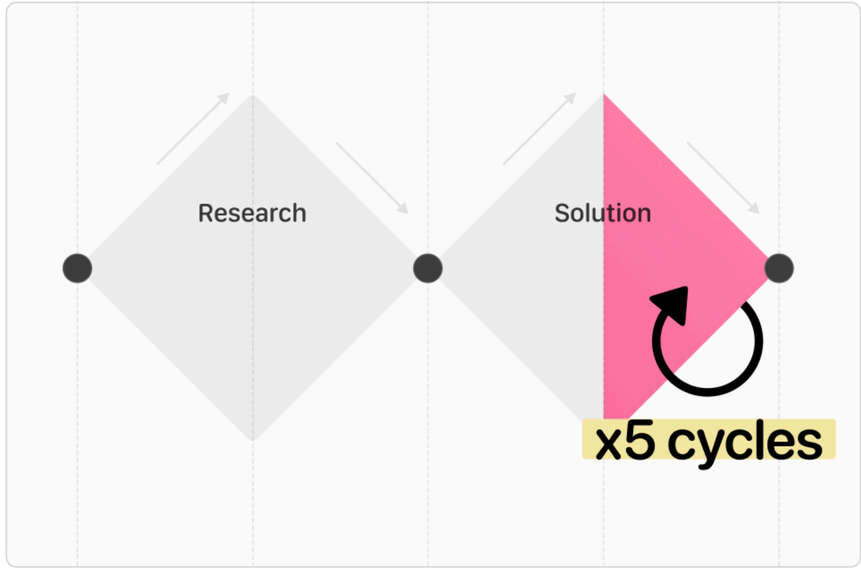
Following the workshop, I conducted deep-dive interviews with leads across flagship AAA titles. This revealed the project's true organizational complexity: 6 separate departments needed to collaborate across fragmented toolchains (combining publishing tools, task trackers, and community hubs).

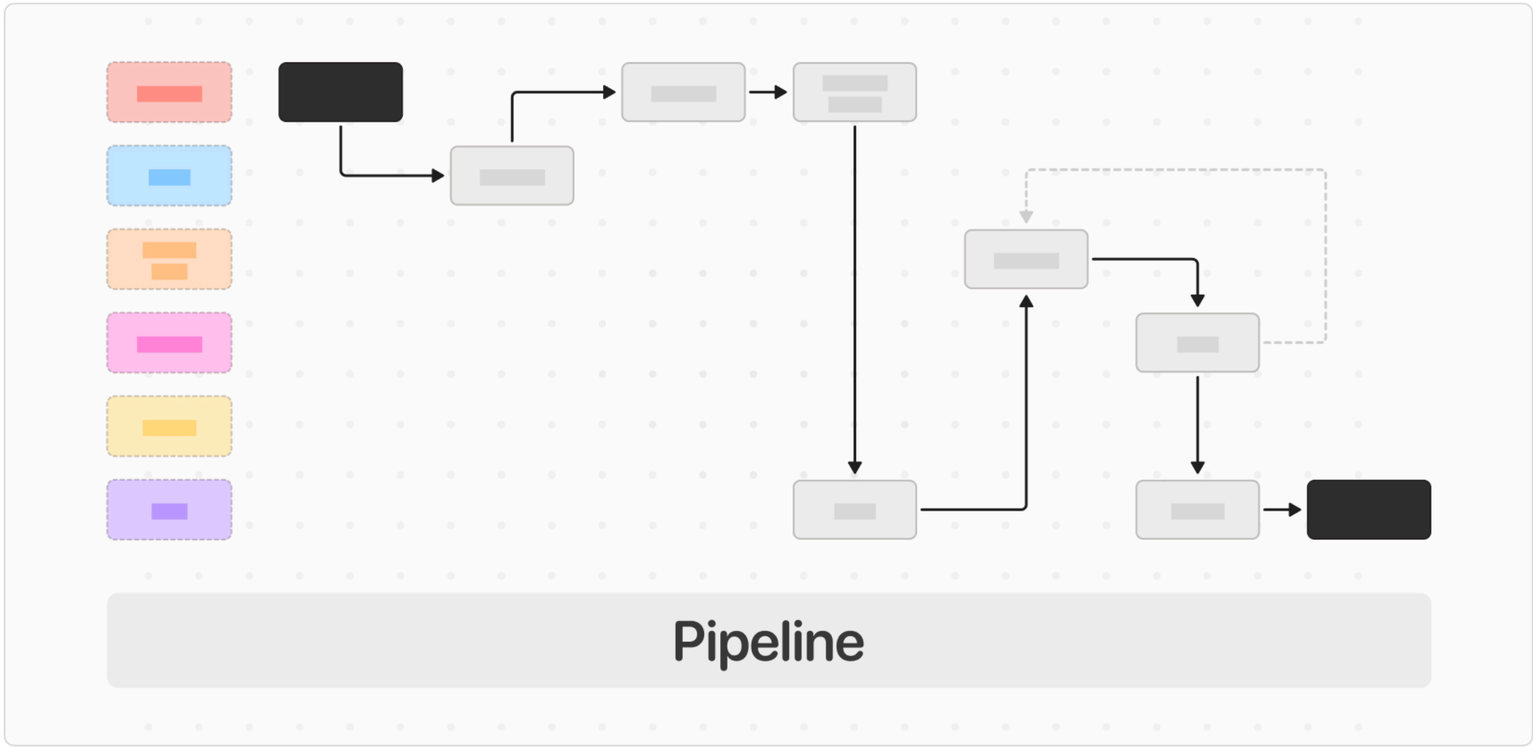
Across 5 iterative cycles, I engineered the workflow to introduce mandatory Brand Safety validation gates while decoupling global asset vetting from local market publishing.

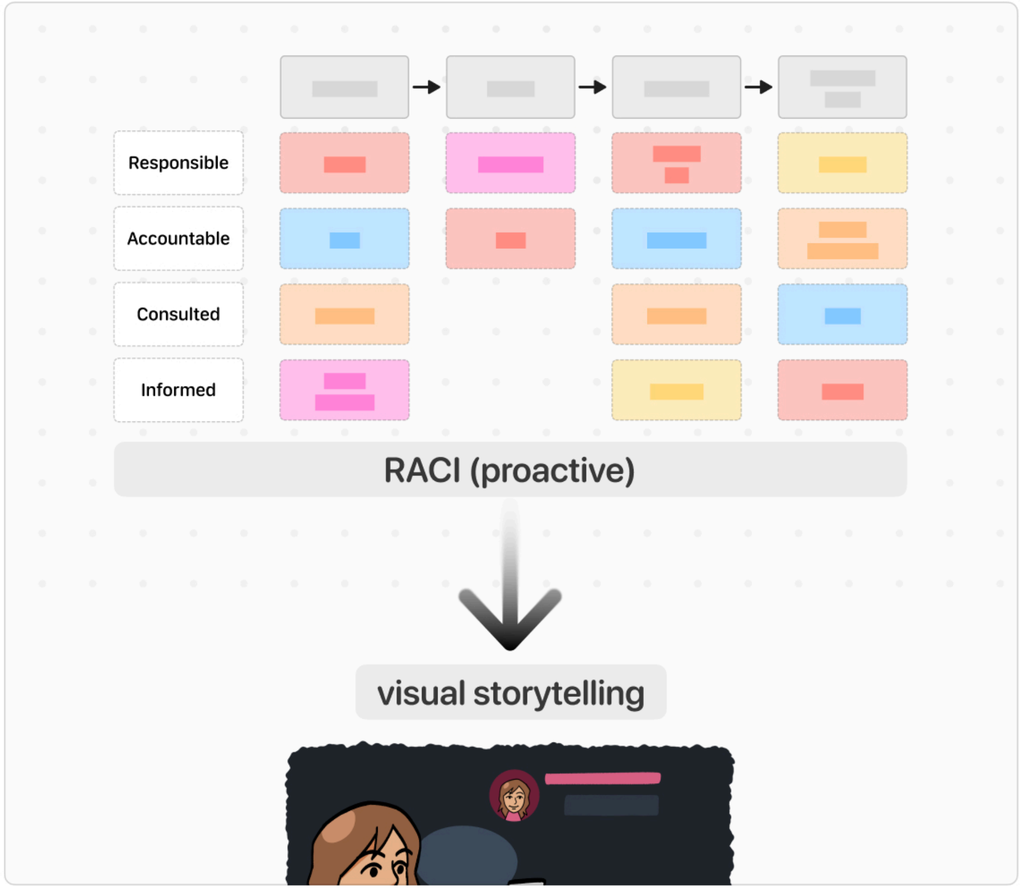
Large-Scale Validation

Once the internal workflow was established, I designed a quantitative survey framework for the 2,000+ Discord creator community to verify creator interest, receptivity, and buy-in prior to live rollout.









RACI Governance

To turn this multi-department workflow into clear enterprise governance, I structured the process into an operational RACI framework. I defined two distinct execution pathways: a Proactive model for planned campaign activations and a Reactive model for spontaneous community clips.

Persuasive Storytelling

To secure executive buy-in, I translated these models into visual storyboards. By mapping the end-to-end human journey (from a creator's Discord submission through Brand Safety approval to local channel publishing), these visuals transformed complex technical handoffs into an engaging, accessible story.

This narrative approach secured 100% executive approval across all 3 key department leads, delivering a production-ready operational blueprint for trial rollout.

100%

Executive Approval

3

Approving Executive Leads

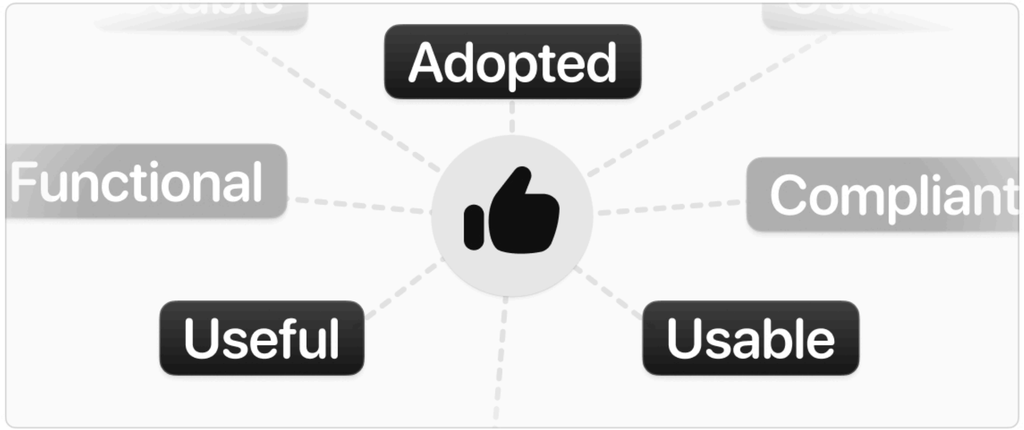
* Excerpt from the executive presentation storyboard (Anonymized)

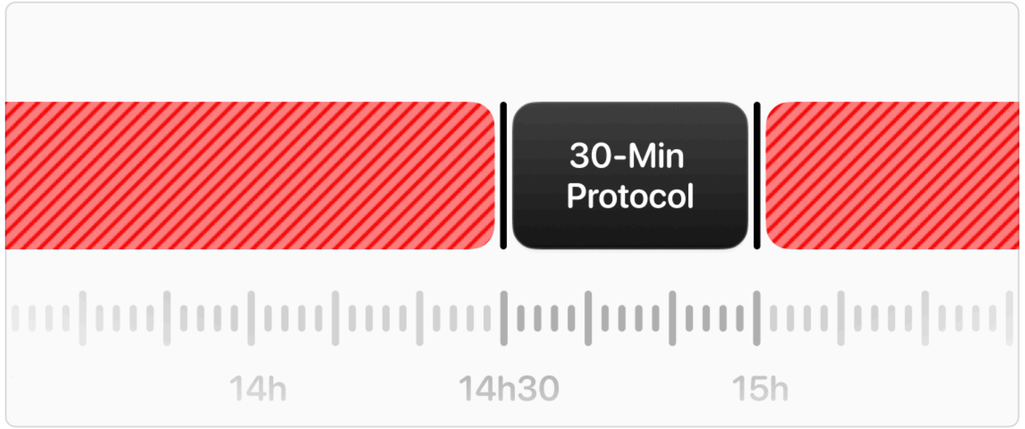
Impact and Reflections

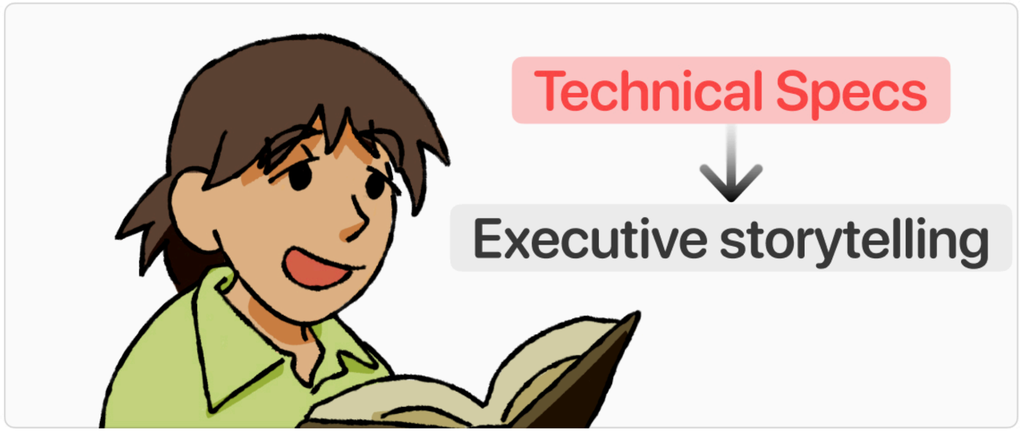
This project reinforced my core conviction that a great solution is not merely usable on paper, but actively adopted and embraced by its operational users.

Navigating strict 30-minute availability limits with internal staff taught me to design highly efficient interview protocols. When third-party dependencies created project downtime, I maintained momentum by executing baseline comparative usability evaluations for an internal creator administration platform.

Ultimately, it proved that visual storytelling is one of the most effective ways to bridge technical systems and business decision-makers, turning complex operational proposals into compelling, buy-in-ready initiatives.

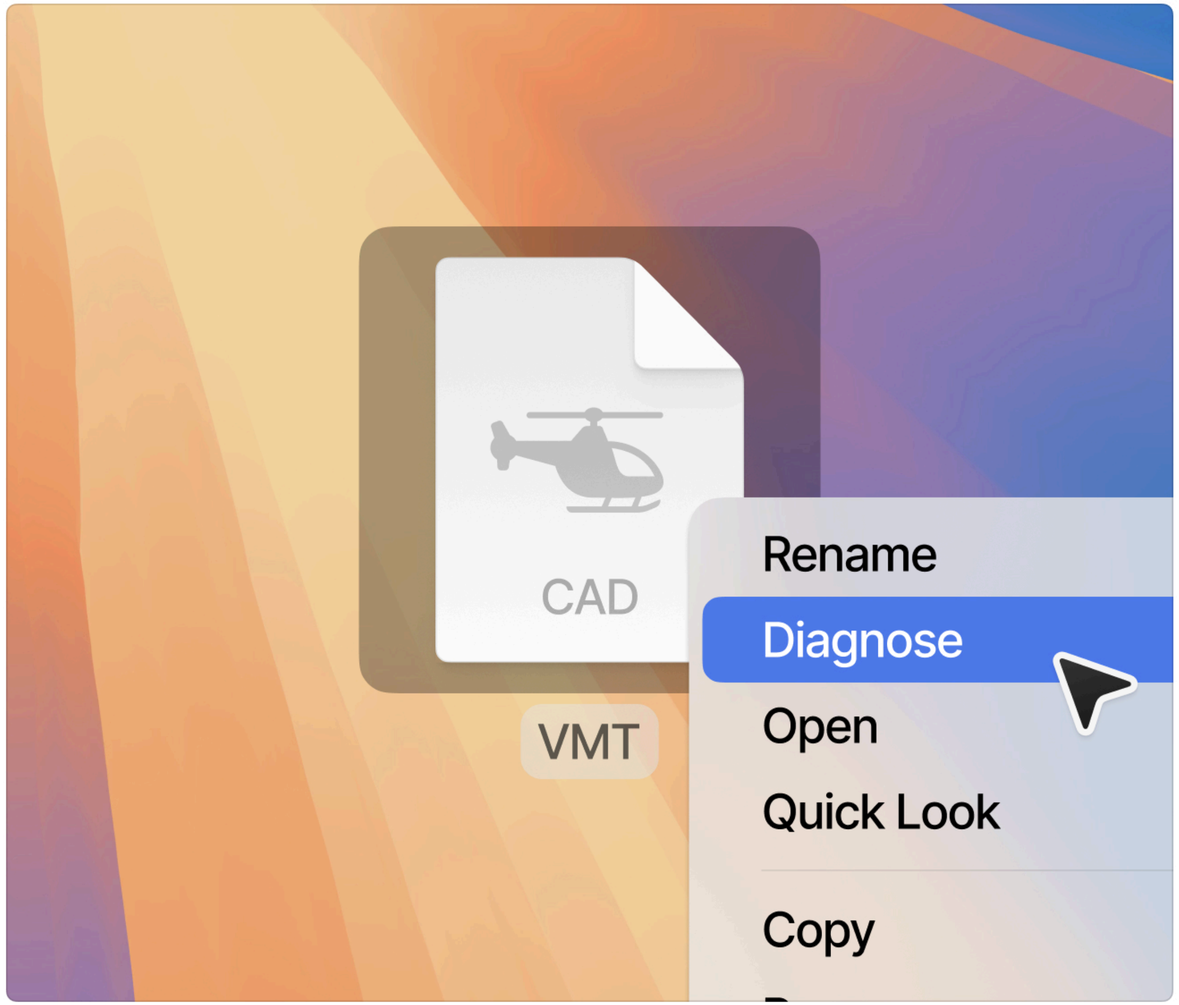






13 **Case Study 2: VMT** Optimizing Cognitive Affordance in Industrial CAD Training

IMS & Airbus Lab, May 2025 - Jul 2025, Bordeaux



Actual VMT software interfaces are withheld to respect client IP. All visuals are generic reconstructions built to demonstrate UX methodology.

Role: UX Consultant & Cognitive Ergonomist Intern

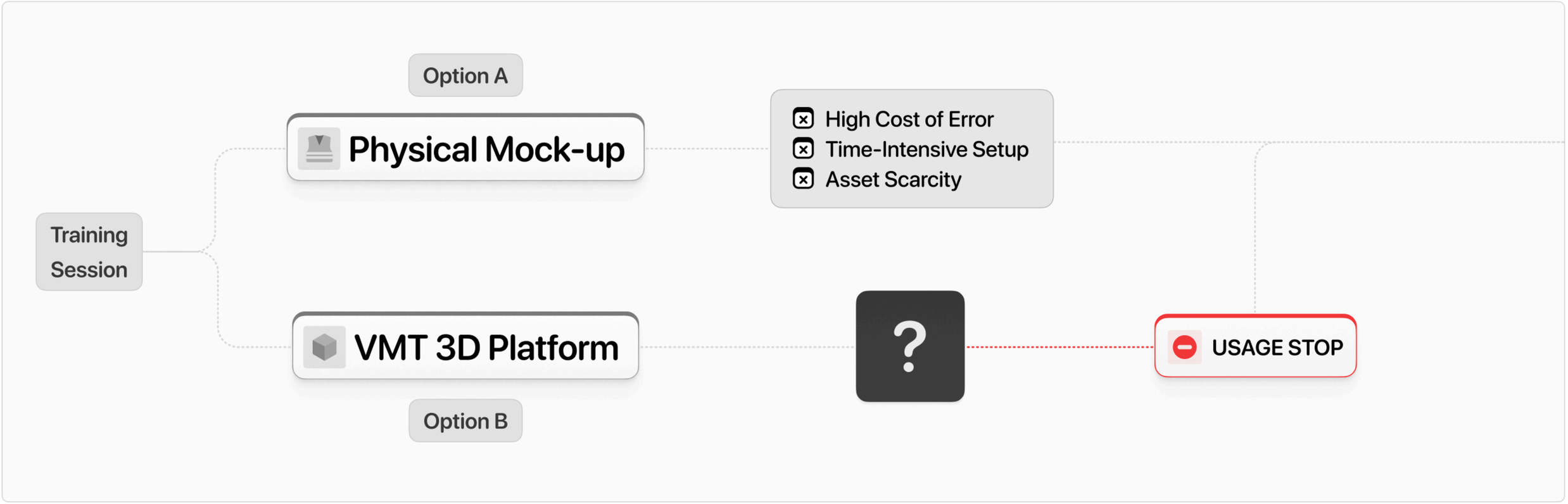
Domain: Industrial CAD Software, Virtual Reality Simulation & Ergonomics

Context

When Airbus Helicopters built the Virtual Maintenance Trainer, a 3D simulation platform designed to train aeronautical technicians on complex maintenance procedures, they created a tool with high-fidelity technical modeling that ultimately struggled to gain traction in actual classrooms.

Diagnostic Challenge

The core problem was a total breakdown in adoption. Instructors were actively bypassing the digital simulator during live lessons, falling back on physical mockups or paper manuals. Because physical aviation mockups are scarce and expensive to maintain, abandoning the digital platform created severe training bottlenecks around technician certification timelines.

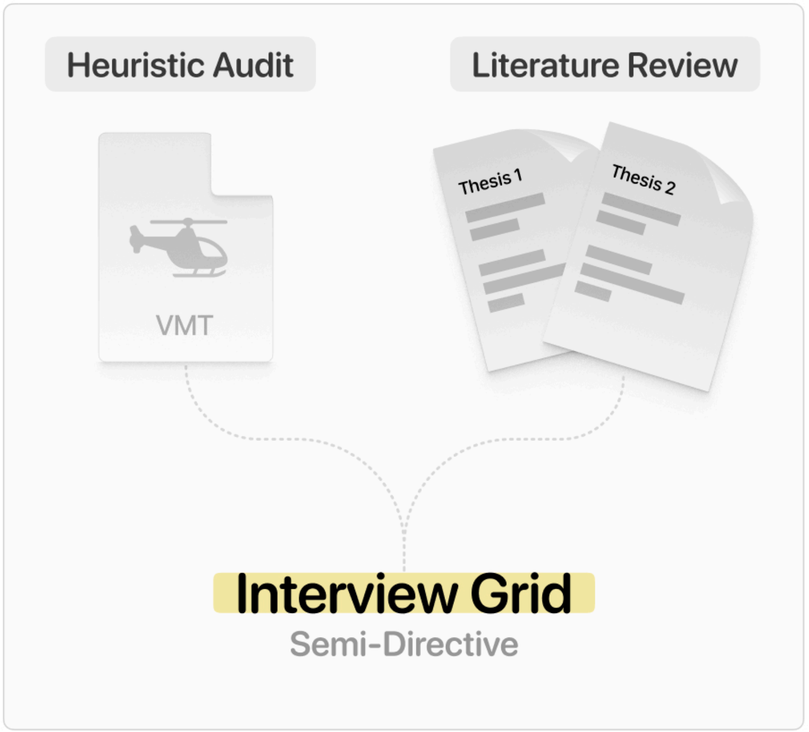


Scope

Operating as an embedded UX Consultant through the IMS Laboratory, my goal was to uncover the cognitive and ergonomic root causes behind this resistance and realign the interface with actual classroom workflows.

Discovery

To immerse myself in the environment and understand the software's baseline operation, I conducted an initial heuristic walkthrough to identify purely technical friction points, such as basic navigational dead ends. Alongside reviewing three aeronautical doctoral theses on simulation learning, this initial audit allowed me to build a highly targeted, semi-directive interview grid tailored to the Airbus training context.



Qualitative Research

During this early phase, a key insight emerged: B2 Avionics instructors adapted to the software with ease, whereas B1.3 Mechanical instructors rejected it entirely because the digital environment offered zero tactile feedback for physical engine parts. Armed with my interview grid, I headed to Aérocampus Aquitaine to conduct seven in-depth interviews with instructors and trainees, processing the audio rapidly via WhisperAI.

Field Discoveries

When observing live classes, I noted that instructors naturally gravitated toward physical mockups rather than opening the 3D software. I pivoted my methodology toward informal contextual inquiries between sessions, which revealed that the navigational overhead of setting up the software severely disrupted their instructional pacing.

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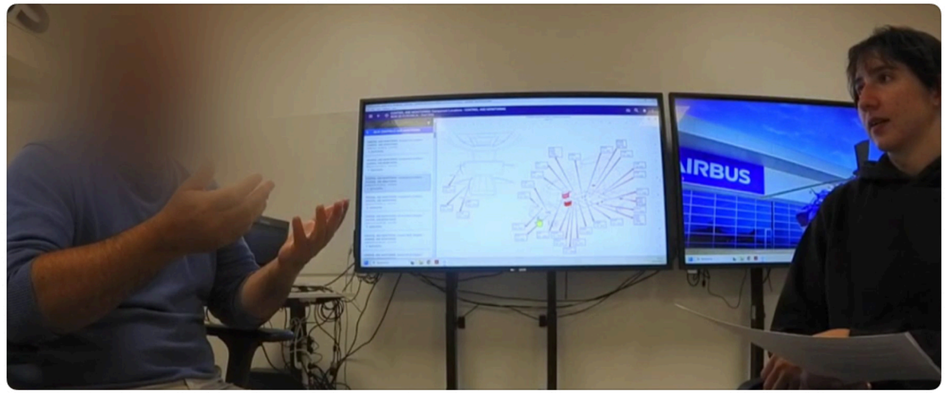
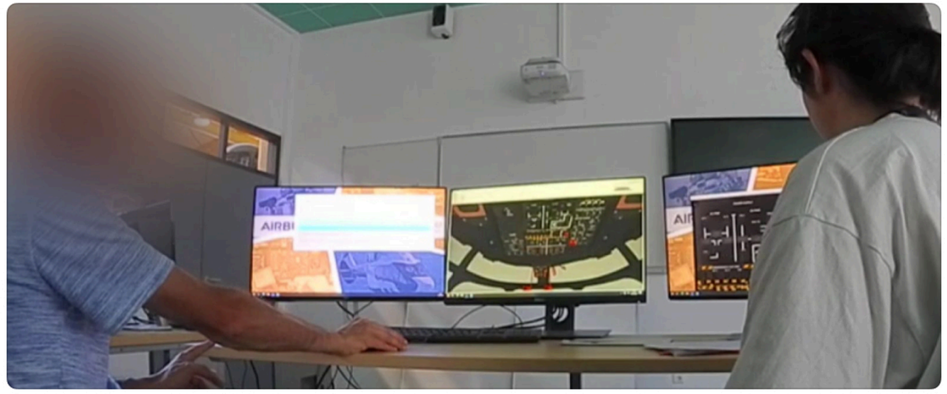
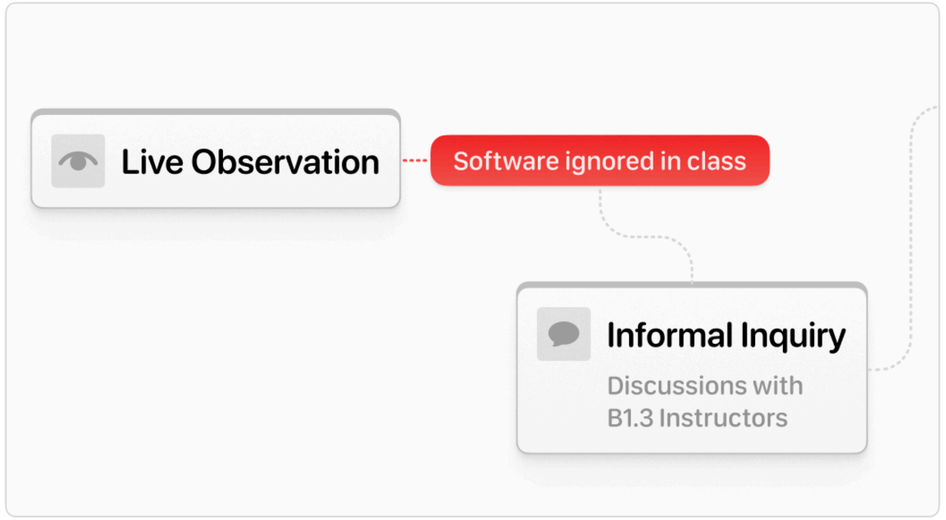
B1.3 Instructor Interviews

2

Trainee Interviews



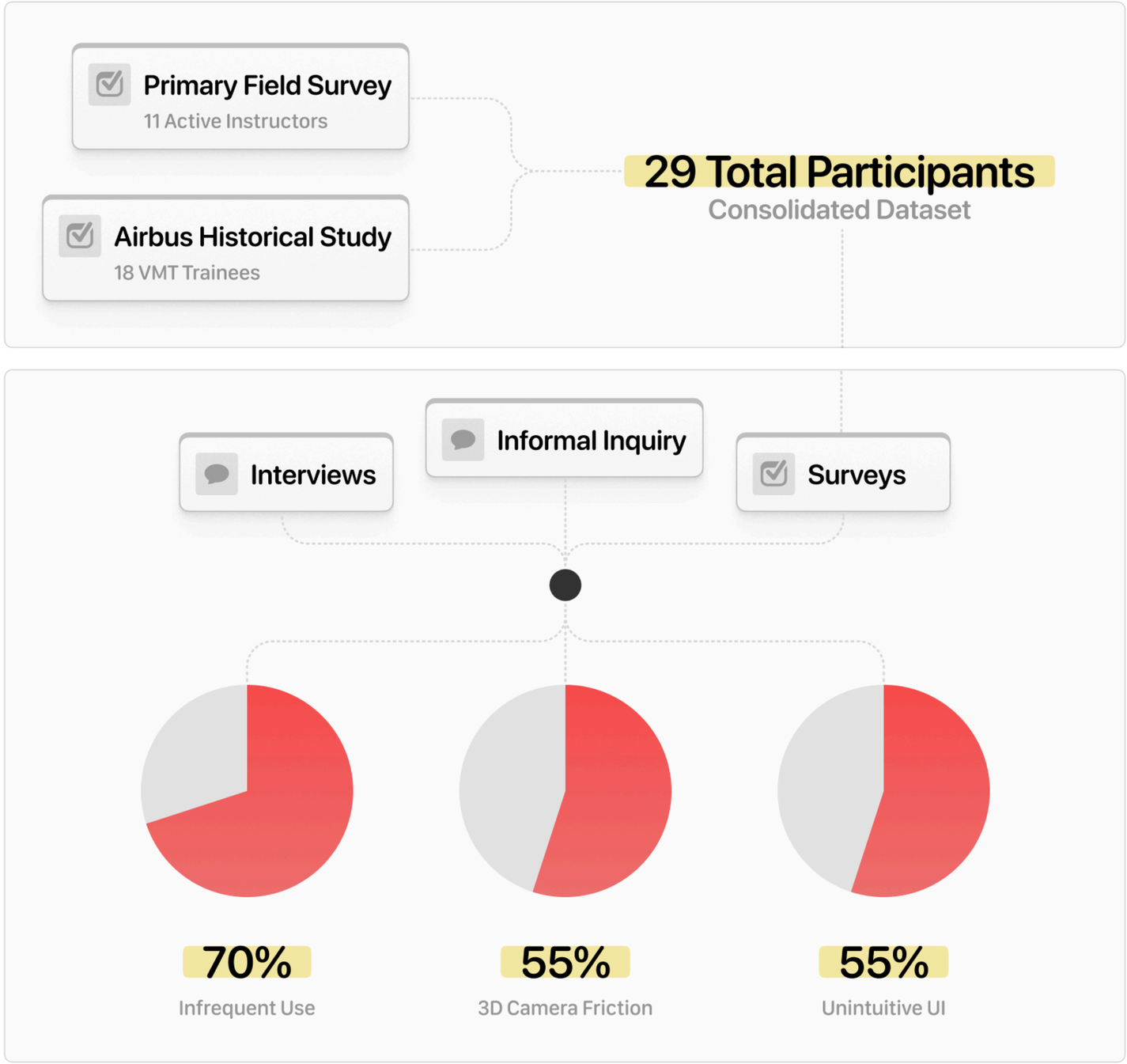




Quantitative Triangulation

To back up these qualitative insights with hard data, I deployed a primary survey to active instructors to measure system usability and task friction. I then triangulated my 11 primary survey responses with a historical internal Airbus study of 18 VMT users, creating a consolidated and highly relevant dataset of 29 participants.

The data proved the issue was systemic: 70% of instructors used the software less than once a month, and 55% cited 3D camera movement and interface clarity as major blockers.



Heuristic Synthesis

With my raw data collected, I needed a way to objectively classify my findings. I adopted the 10-Point Ergonomic Heuristic Framework by Colombo and Pasch to synthesize the interview and survey data, evaluating vectors like Ergonomic Transparency and Cognitive Load.

Usability Diagnosis

This structural analysis allowed me to map a total of 28 distinct usability issues across the platform. From this comprehensive diagnosis, I engineered 25 concrete ergonomic recommendations, ranging from architectural menu reorganizations to permissive interaction models. To keep this case study focused, I have isolated the top three most severe cognitive friction points:

- Unstandardized camera navigation disoriented users during engine inspections.
- Nested menus constantly forced instructors to look away from the task while lecturing.
- Selecting small 3D engine components required pinpoint mouse accuracy with zero visual hover affordance.

Unstandardized camera navigation

Nested menus

Imprecise selection

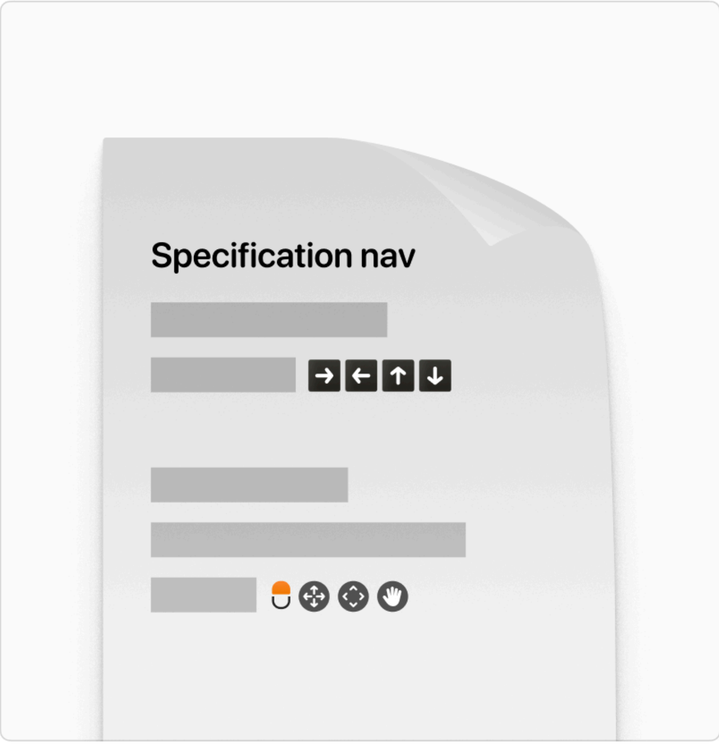
Interaction Specification

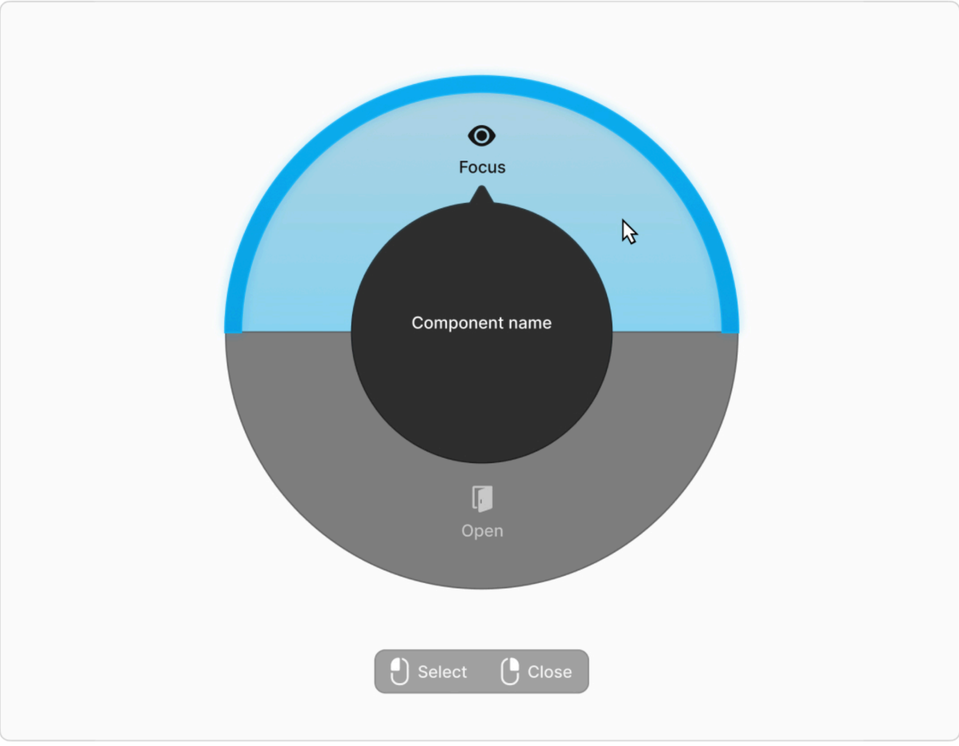
Because my mandate focused on interaction architecture rather than high-fidelity visual assets, I developed ergonomic interaction specifications to solve these exact issues.

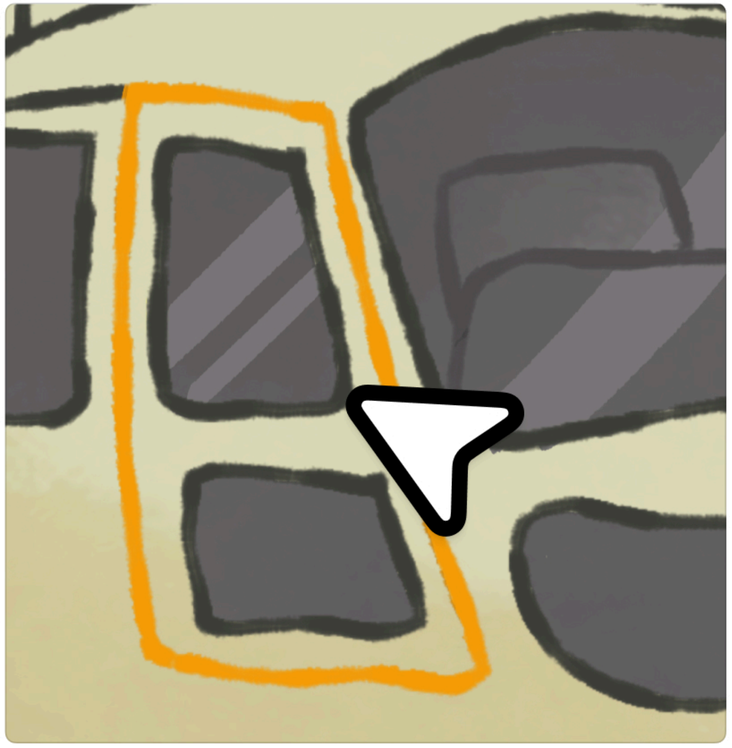
I proposed a dual-mode camera system combining standard WASD movement for navigating rooms with automatic orbit controls centered on selected components.

To reduce menu dependency, I designed a cursor-anchored radial menu leveraging Fitts's Law to give instant access to key actions.

Finally, I specified permissive click tolerances, dynamic component outlining, and structured error messages to eliminate guesswork.







Impact and Reflections

The final deliverables included a 63-page comprehensive UX analysis report detailing all 28 usability blockers and their corresponding 25 technical specifications. To ensure actionable stakeholder buy-in, I distilled this exhaustive report into an executive slide presentation that pitched the six most impactful strategic ideas.

This specification package (bridging the comprehensive research report and the focused executive pitch) received unanimous sign-off from engineering leads at Airbus Helicopters and Telespazio for integration into production.

Looking back, pivoting away from formal observations to capture contextual setups proved that even highly detailed technical simulations fail if the underlying interaction design ignores human cognitive limits.

25

UX Specs Delivered

63 Pages

Comprehensive UX Analysis Report

Case Study 3: EmCo Decoupling Emotional State from Cognitive Workload

ENSC, Jan 2026, Bordeaux, 5-Day Cross-Disciplinary Sprint

Role: Lead UX Designer & Systems Architect (Academic Sprint)

Domain: MedTech, Biometric Telemetry & Cognitive Ergonomics

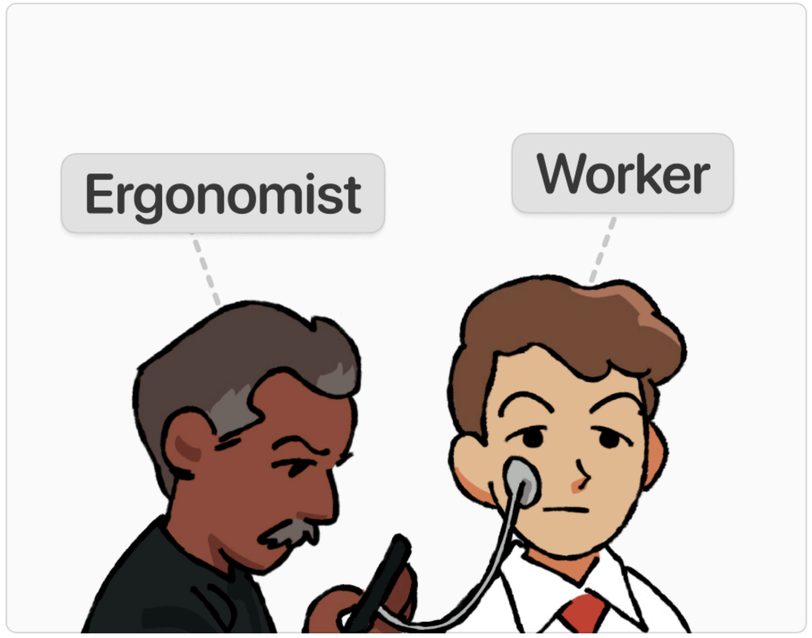
Context

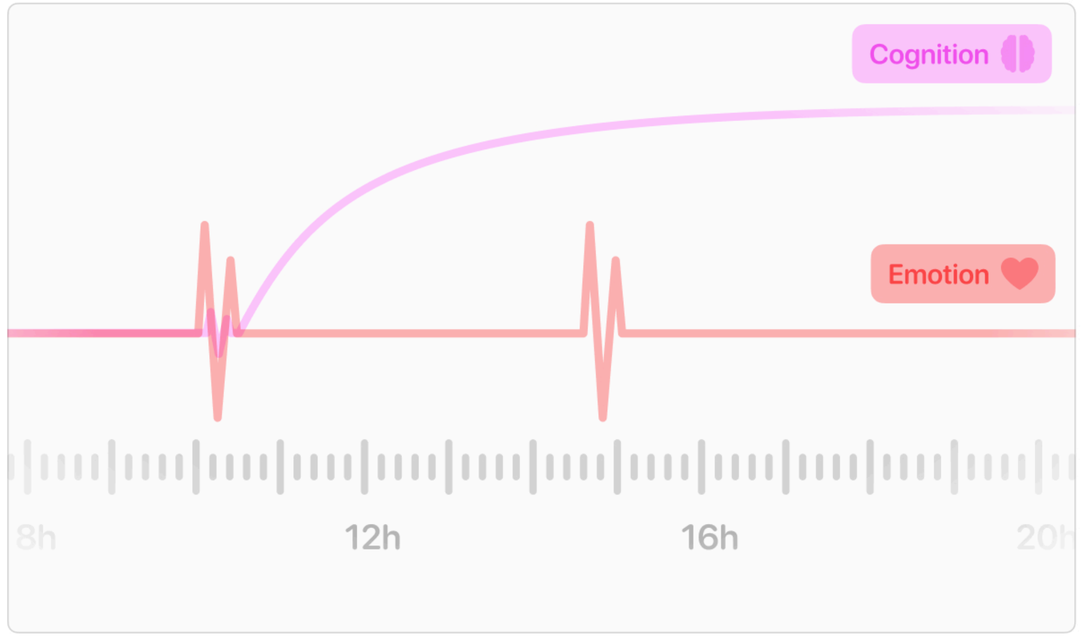
Corporate ergonomists evaluate workplace quality of life and prevent psychosocial risks, but traditional methods rely heavily on retrospective, subjective questionnaires. The EmCo project aimed to create an objective diagnostic system translating non-invasive biometric telemetry into actionable Quality of Work Life (QVT) indicators.

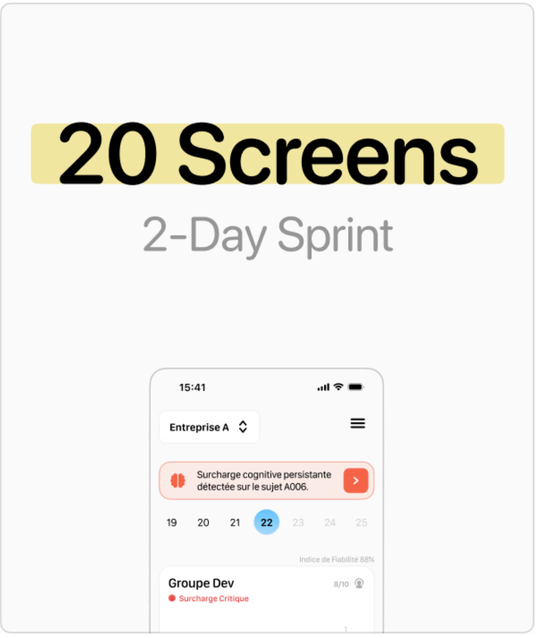
Scientific Challenge

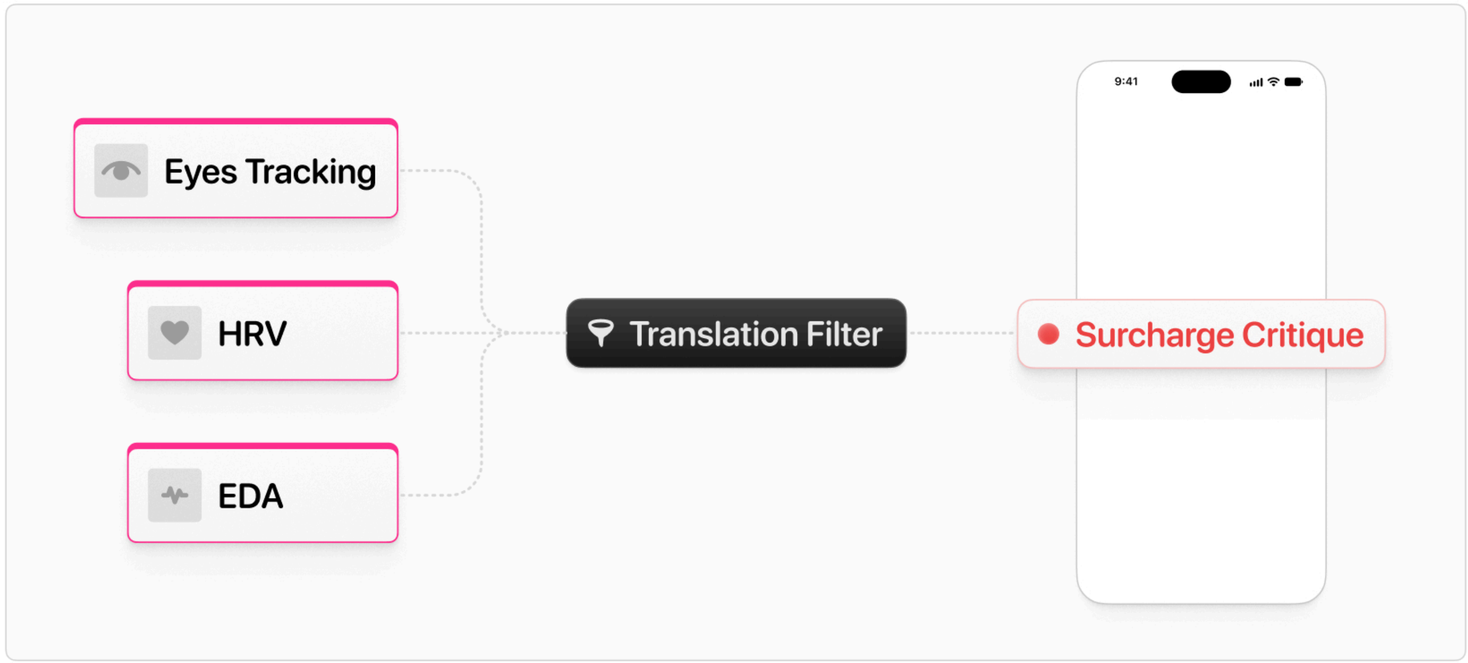
Emotion and cognition activate similar physiological responses but operate on fundamentally different temporal scales: emotion manifests as sudden, brief physiological spikes, whereas cognition represents sustained resource mobilization. The core UX challenge was visually decoupling these biological realities without overwhelming the user with raw data.

As the UX Lead across a six-person cross-disciplinary team, I owned the information architecture and technical UI decisions, tasked with delivering a complete 20-screen mobile design system.







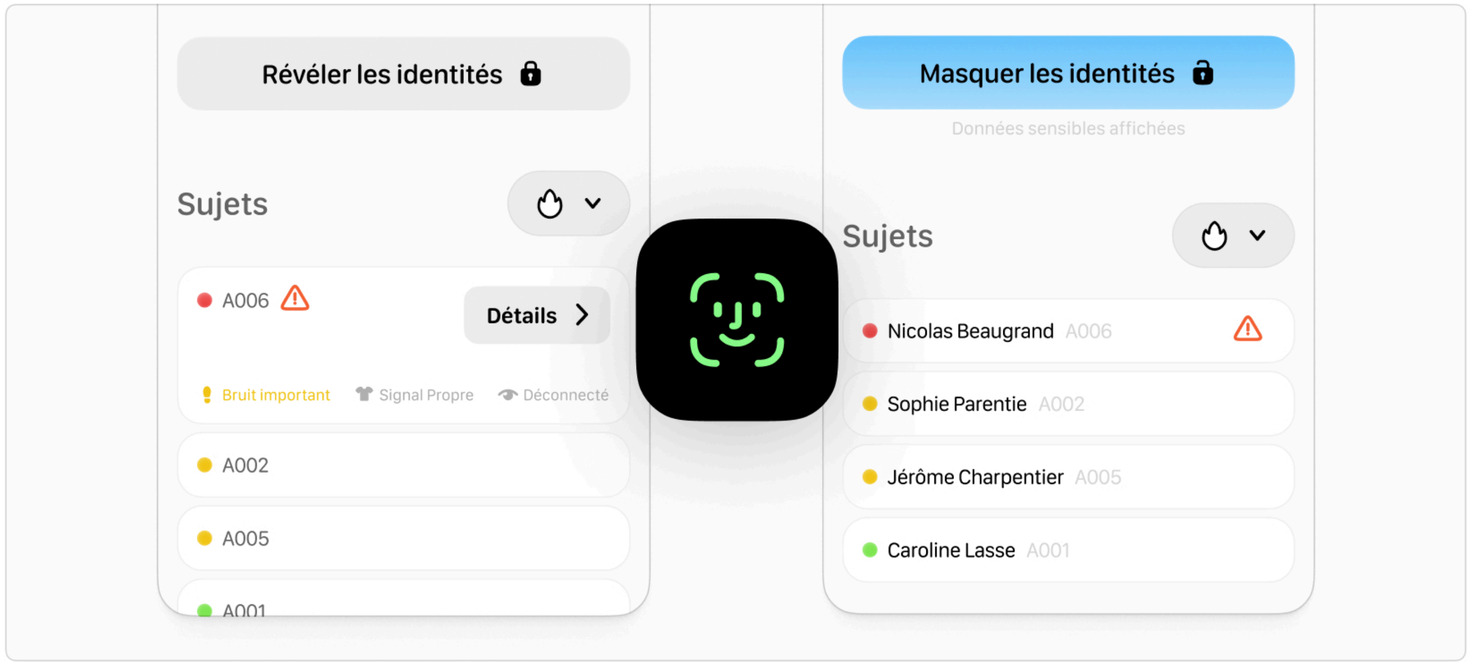


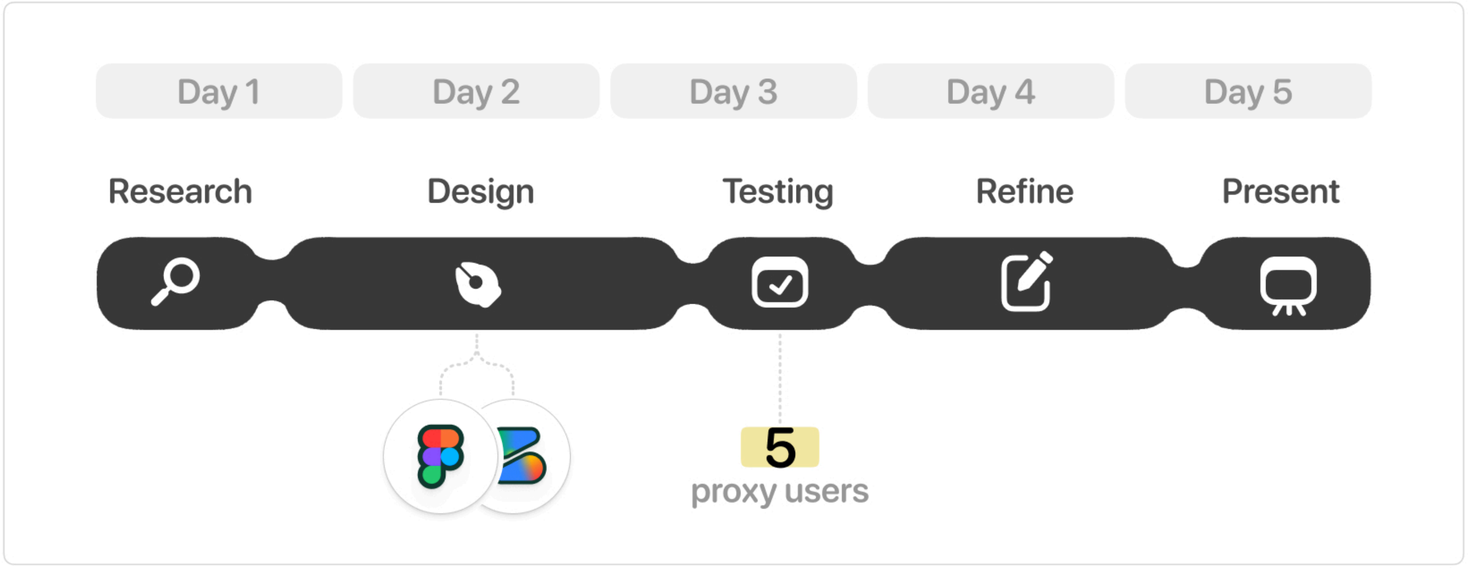
Data Translation

Raw data is useless without interpretation. To prevent cognitive overload, I translated complex biometric streams (HRV, EDA) into actionable, human-readable status labels like "Focus Productif" and "Surcharge Critique".

Privacy

To handle privacy seamlessly during live monitoring, I implemented iOS-style FaceID; this allowed ergonomists to securely and repeatedly lift patient anonymity for clinical checks without the long-term interaction friction of passwords.



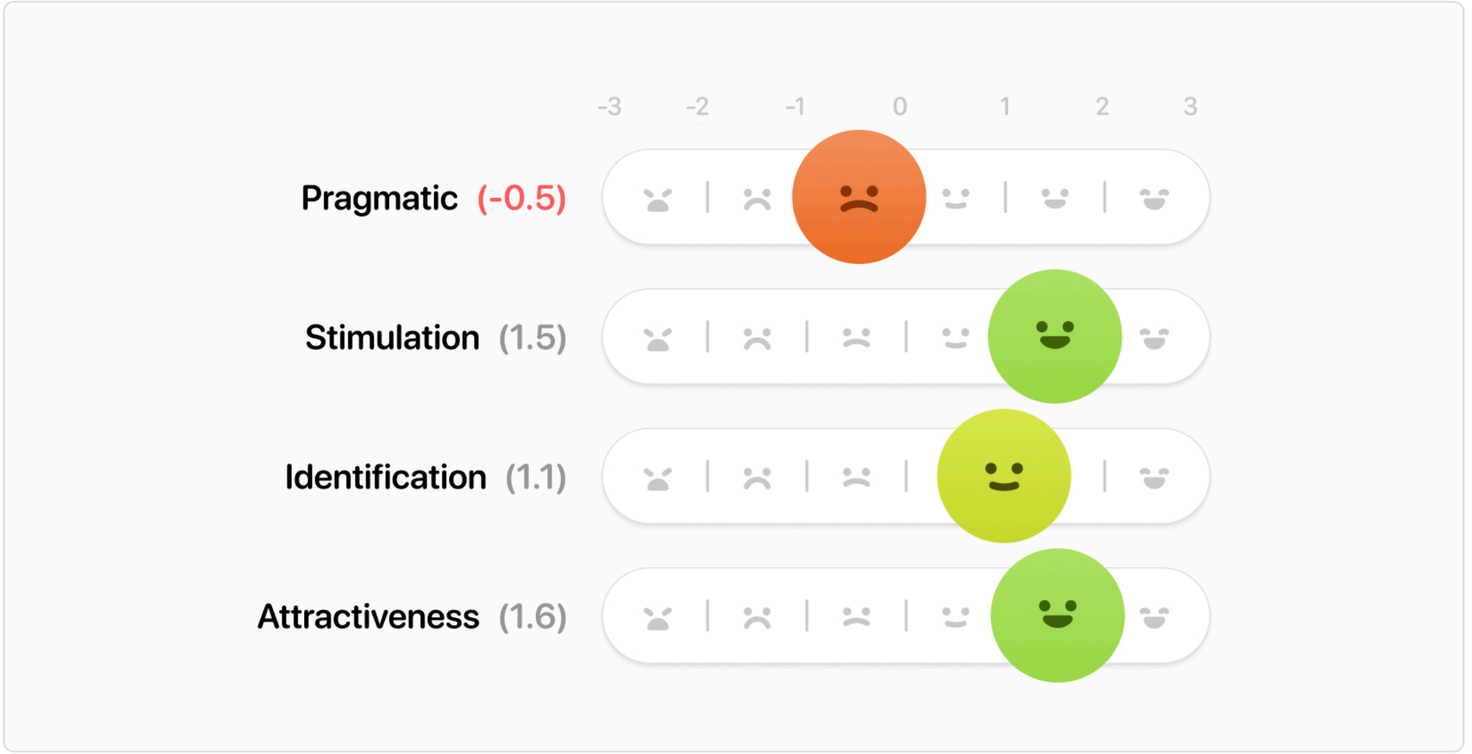


Rapid Execution

Operating within a strict 5-day conceptual sprint, I aggressively timeboxed the high-fidelity UI execution to just 48 hours. To maximize efficiency, I utilized AI Studio to audit wireframe layouts before executing Day 3 usability testing with 5 cognitive engineering professors and PhD researchers acting as proxy users.

Usability Evaluation

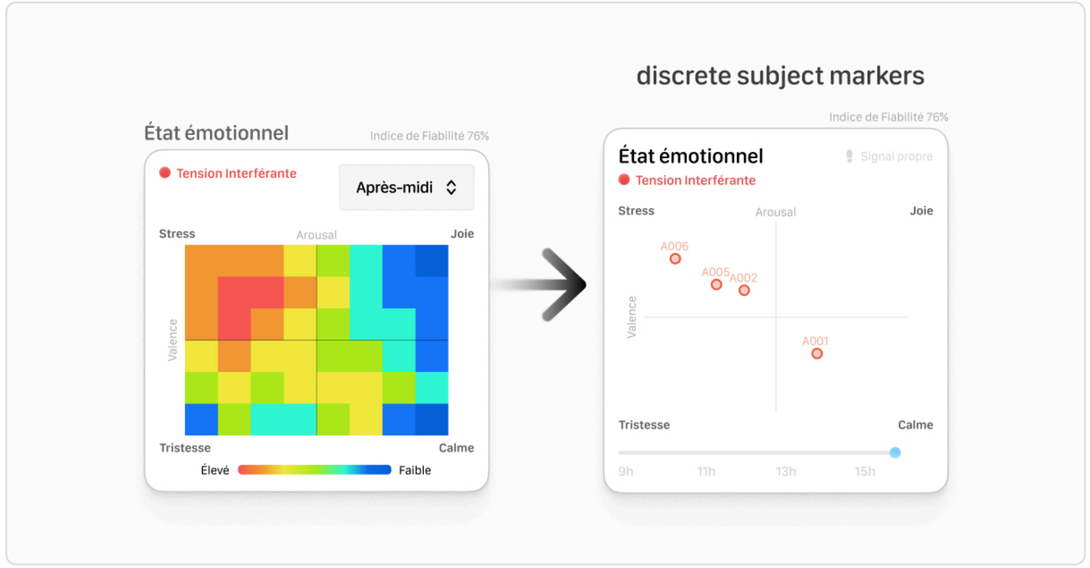
AttrakDiff testing revealed high engagement (+1.5 Stimulation) but highlighted a critical usability flaw (-0.5 Pragmatic Quality), as continuous 3D emotional heatmaps proved too dense to interpret during extended monitoring.

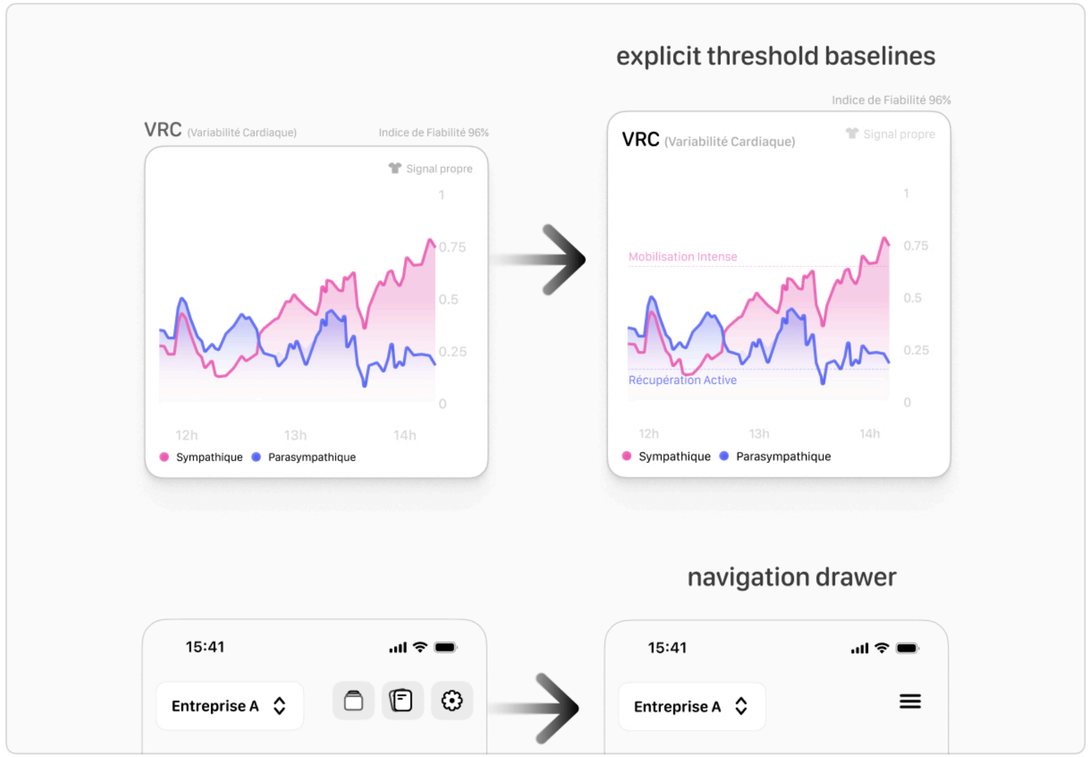


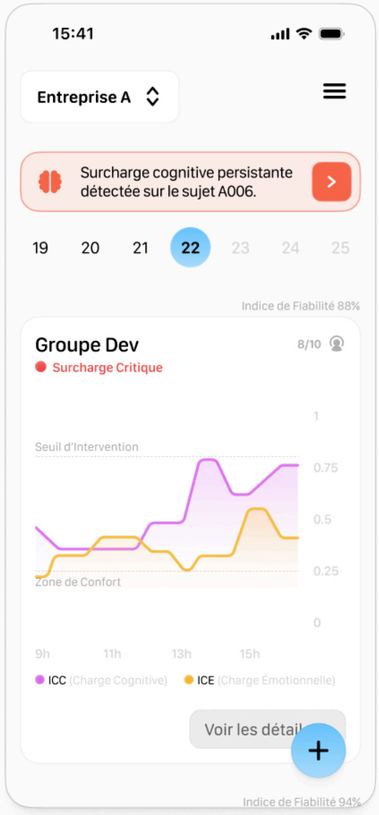
Empirical Iteration

To eliminate this cognitive friction, I executed three targeted architectural changes to aggressively filter visual noise:

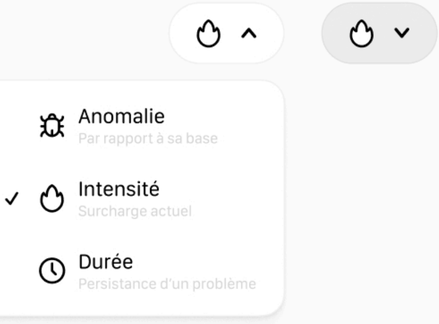
- I abandoned the dense heatmaps in favor of a flattened grid based on Russell's model. By plotting discrete subject markers, the eye is instantly guided to critical outliers rather than scanning a chaotic 3D map.
- I paired this grid with a temporal scrubbing slider, empowering users to track short emotional peaks over time and isolate them from baseline cognitive fatigue.
- I added explicit threshold baselines across all cognitive and biometric charts to remove interpretive guesswork, and consolidated ambient header actions into a labeled hamburger menu to keep the screen focused purely on diagnostic tasks.





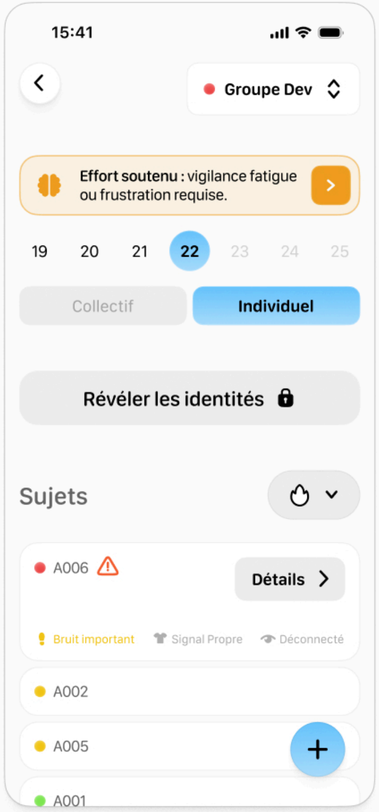


Organizational Overview

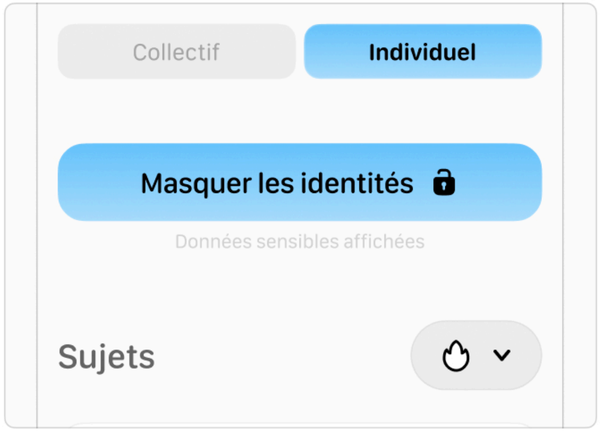


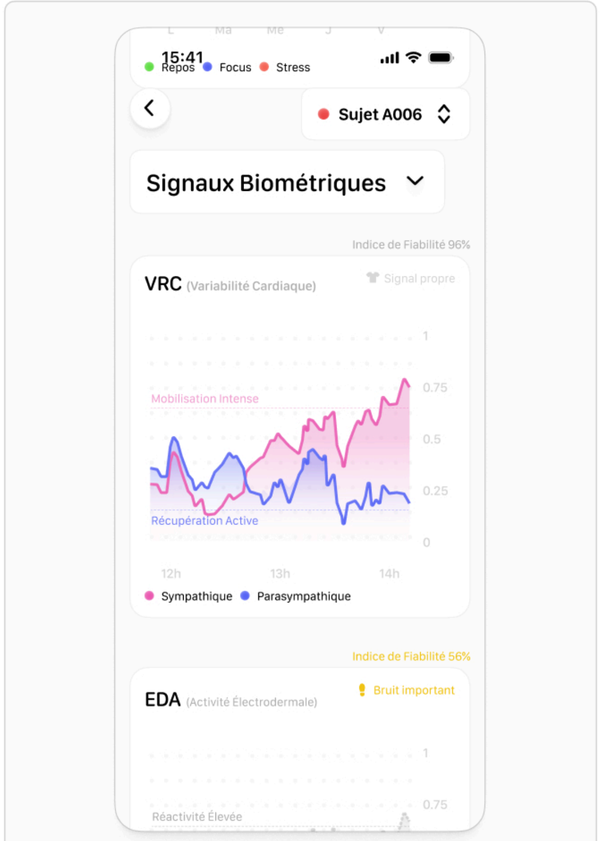
Telemetry sorting & Anomaly filters

* UI microcopy rendered in native French deployment context



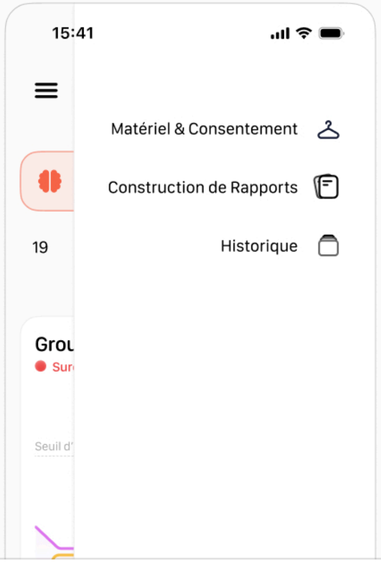
Group Cognitive Workload View





Individual Biometric Telemetry

Hardware & Consent Navigation



Indice de Fiabilité 56%

Groupe Commerciaux

4/6

Tension Sourde

1

Seuil d'Intervention

0.75

0.5

Signal Reliability Index

Cohort vs. Individual View Switcher

15:41

Groupe Dev

15:41

Sujet A006

19 20 21 22 23 24 25

Collectif

Individuel

Masquer les identités

Données sensibles affichées

Sujets

Nicolas

Sophie

Jérôme

Caroline Lasse A001

Anomalie

Par rapport à sa base

Intensité

Surcharge actuel

Durée

Persistance d'un problème

GDPR & Subject Anonymization Controls

Indice de Fiabilité 76%

État émotionnel

Signal propre

Tension Interférante

Stress

Arousal

Joie

A006

A005

A002

A001

Valence

Tristesse

Calme

9h 11h 13h 15h

Group Emotional Valence Chart

Indice de Fiabilité 94%

Bilan de mobilisation temporelle

12h

9h

6h

3h

0

temps cumulé

L Ma Me J V

Repos Focus Stress

Temporal Workload Distribution

Impact and Reflections

Delivering a functional MedTech design system under extreme time limits required aggressive scoping of core user flows. Using AI tools to critique wireframes accelerated iteration loops, while negative pragmatic feedback highlighted the danger of over-designing real-time charts.

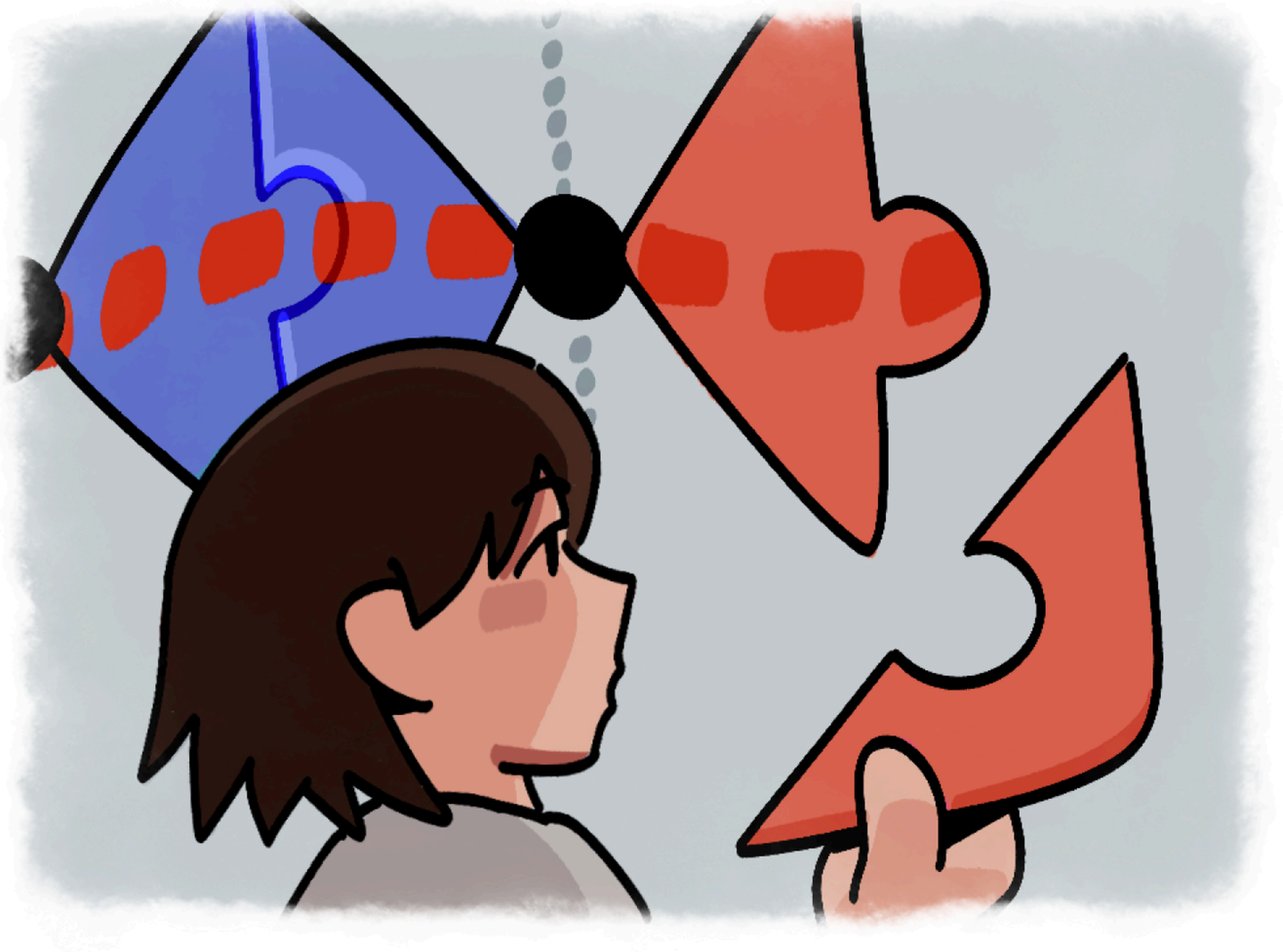
Although the sprint's timeline precluded a second round of testing, the final architectural pivot was designed specifically to let users instantly differentiate acute stress events from baseline recovery. The project culminated in strong validation from the reviewing panel.

If transitioned into a commercial product, the immediate strategic next step would be deploying the iterated prototype in the field: conducting rigorous usability testing with practicing corporate ergonomists to validate the UI changes and refine the QVT indicators for enterprise adoption.

Case Study 4: Multi-Product UCD Sprint

Rapid Systems Architecture & Field Testing

ENSC, Oct 2024 - Feb 2025, Bordeaux

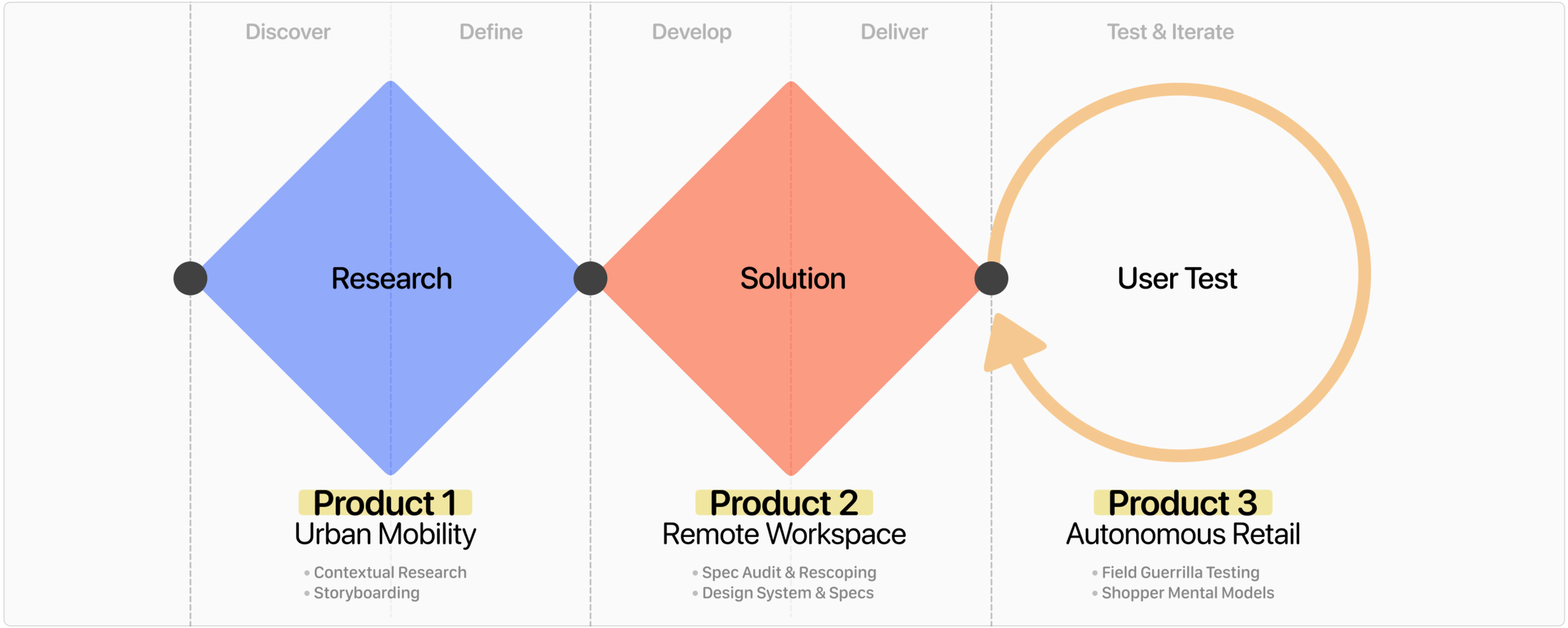


Role: Co-Researcher, Visual Design Lead (DeskFriend) & Testing Co-Lead (Smart Cart) (Academic)

Domain: Urban Mobility, Remote Work Ergonomics & Autonomous Retail

Sprint Context

Following the Double Diamond framework, this intensive multi-product sprint simulated fast-paced agency conditions across three separate product domains. The core objective was to rapidly adapt to unfamiliar industries, conduct targeted contextual research, build modular Figma design systems, and execute empirical field testing under real-world constraints.

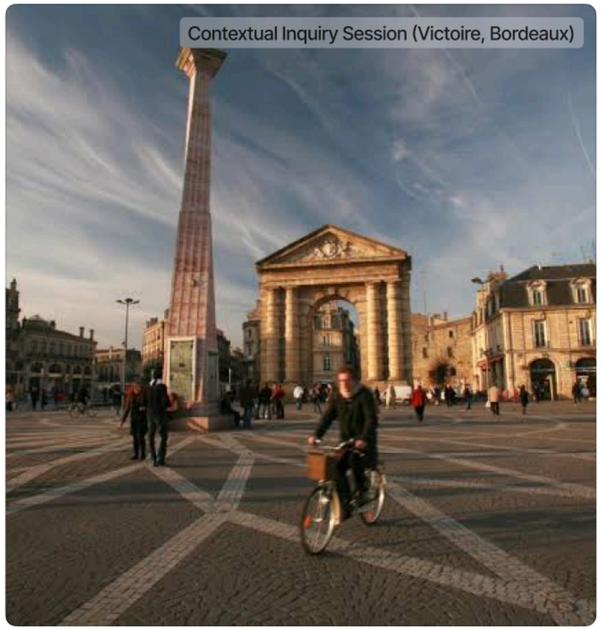


Field Observations

Urban bicycle theft results in 400,000 stolen bikes annually in France. To identify behavioral root causes, we conducted contextual observations and interviews in high-traffic transit hubs across Bordeaux, including Sainte-Catherine Street and Bergonié.

Research Insights

While mapping three distinct personas (the delivery driver, the rushed parent, and the determined ecologist), our research pinpointed micro-stops as the most critical point of vulnerability. The primary friction point was time: the 23.2-second average required to secure a physical U-lock leads cyclists making brief stops to bypass security entirely to maintain their delivery schedules.



23.2-second

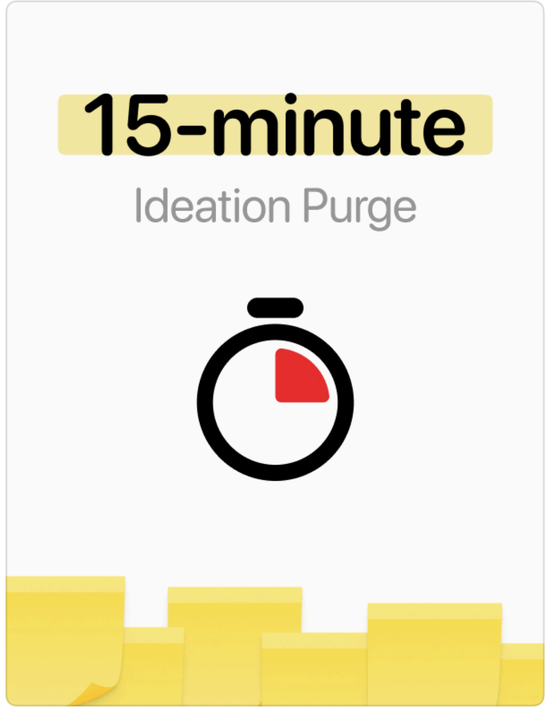
average required to secure a physical U-lock

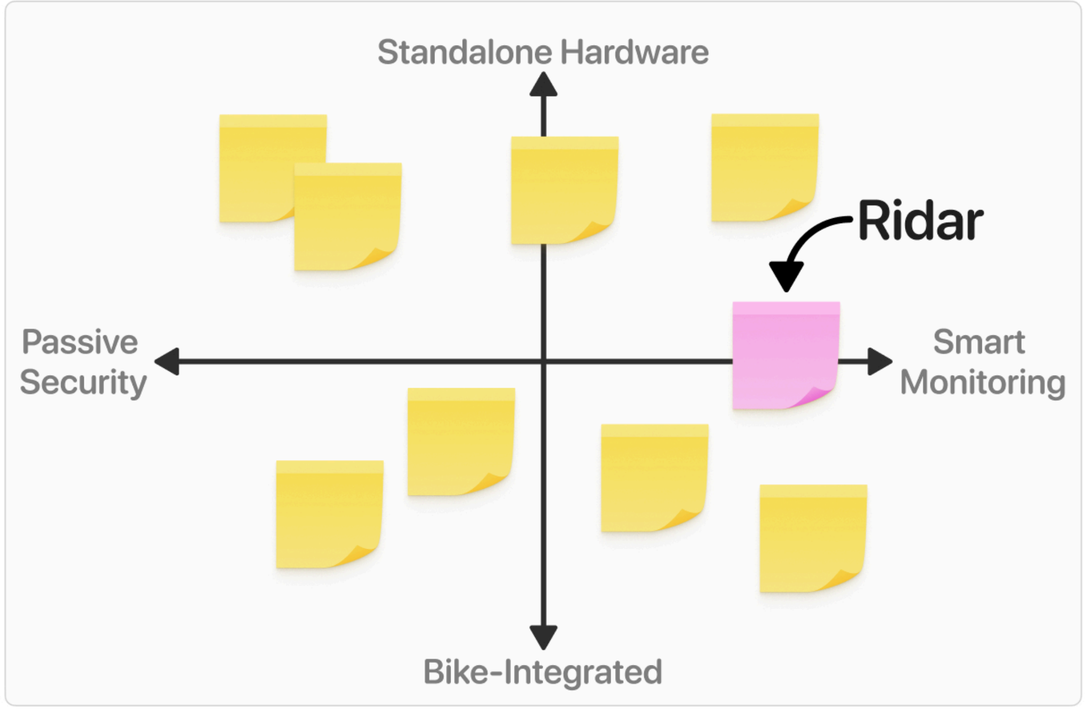
Ideation

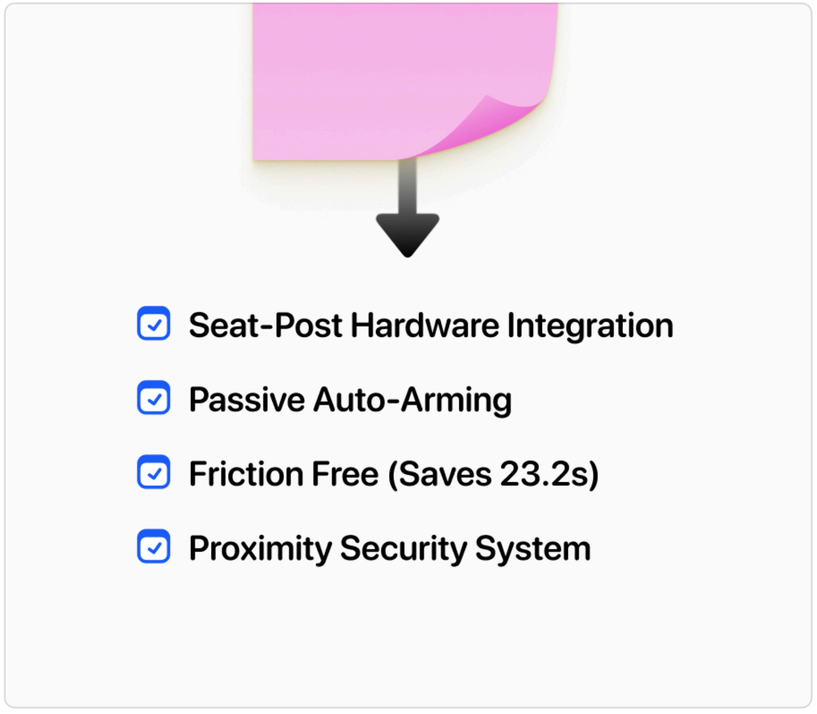
To avoid generic security clichés, our team ran a 15-minute ideation purge to eliminate obvious concepts, leaving a dozen viable ideas. We evaluated these on a matrix structured along two axes: Standalone Hardware vs. Bike-Integrated, and Passive Security vs. Smart Monitoring.

Prioritization

Scoring the surviving concepts on effectiveness, feasibility, and originality birthed Ridar: a phygital security device integrated directly into the seat post. It auto-arms in the background as the cyclist walks away, completely eliminating the 23.2-second physical lock friction.



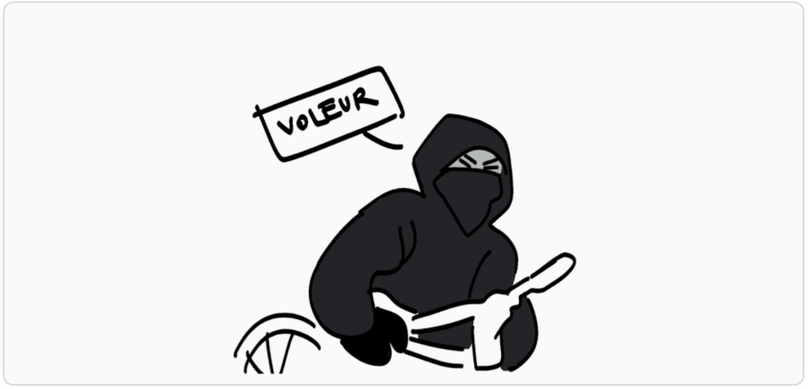




Visual Storyboarding

As Co-Researcher, I illustrated the core user workflow through a storyboard centered on Rayan, a delivery driver persona. The storyboard demonstrates how Ridar passively protects his bike without requiring manual app interaction during quick pick-ups. The system issues instant smartphone alerts and triggers a localized physical alarm upon detecting unauthorized motion, protecting his primary source of income without compromising his delivery speed.

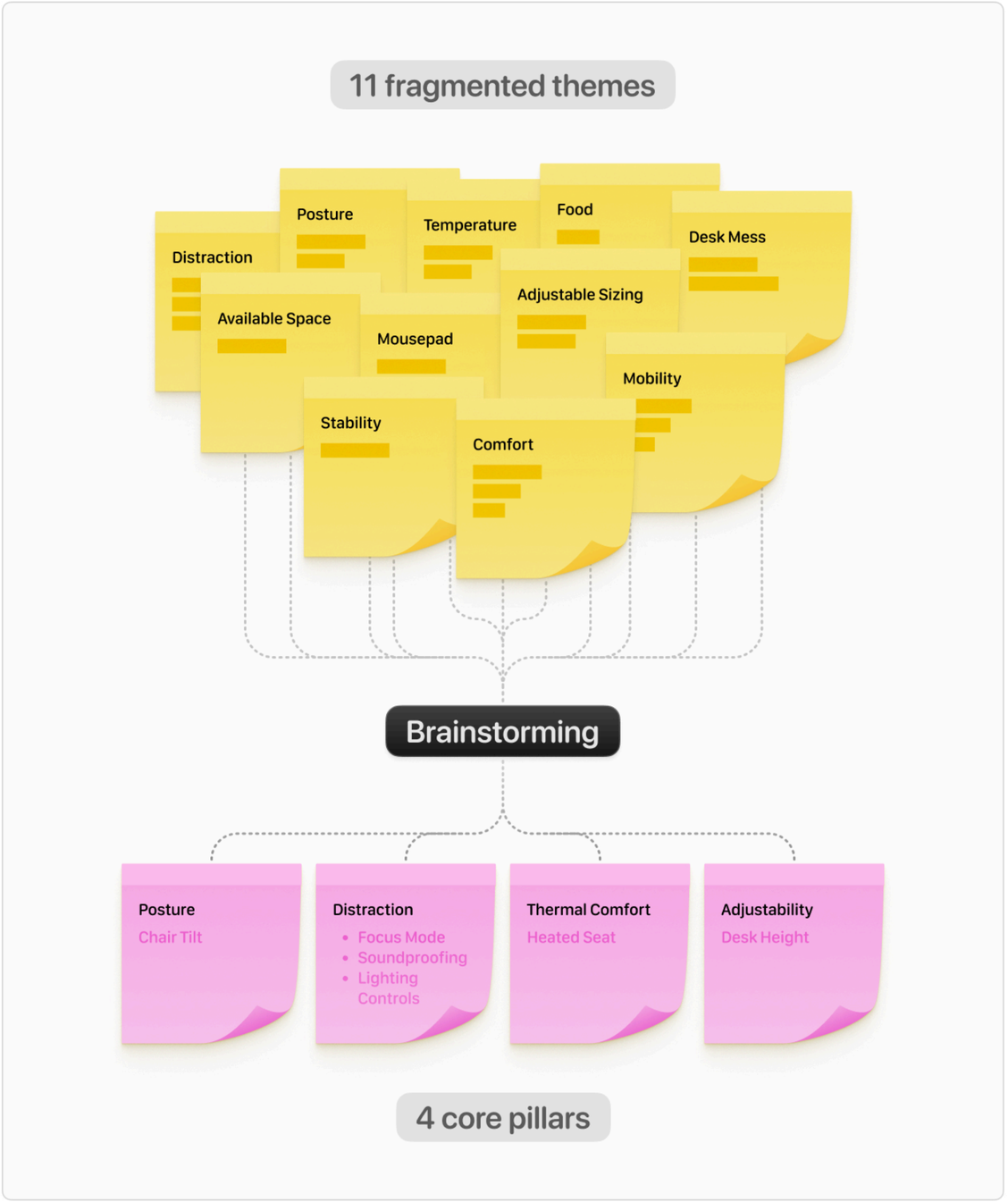
* Excerpt from the User Story (storyboard)



Specification Audit Rescoping

Transitioning to workspace ergonomics, our team inherited an external student spec sheet for a smart desk controller. The initial proposal suffered from severe feature bloat, presenting 11 fragmented themes (some packing up to three sub-features each).

Rather than executing flawed requirements, we held a structured brainstorming session to rationally rescope the project, consolidating the scattered ideas into 4 core UX pillars.



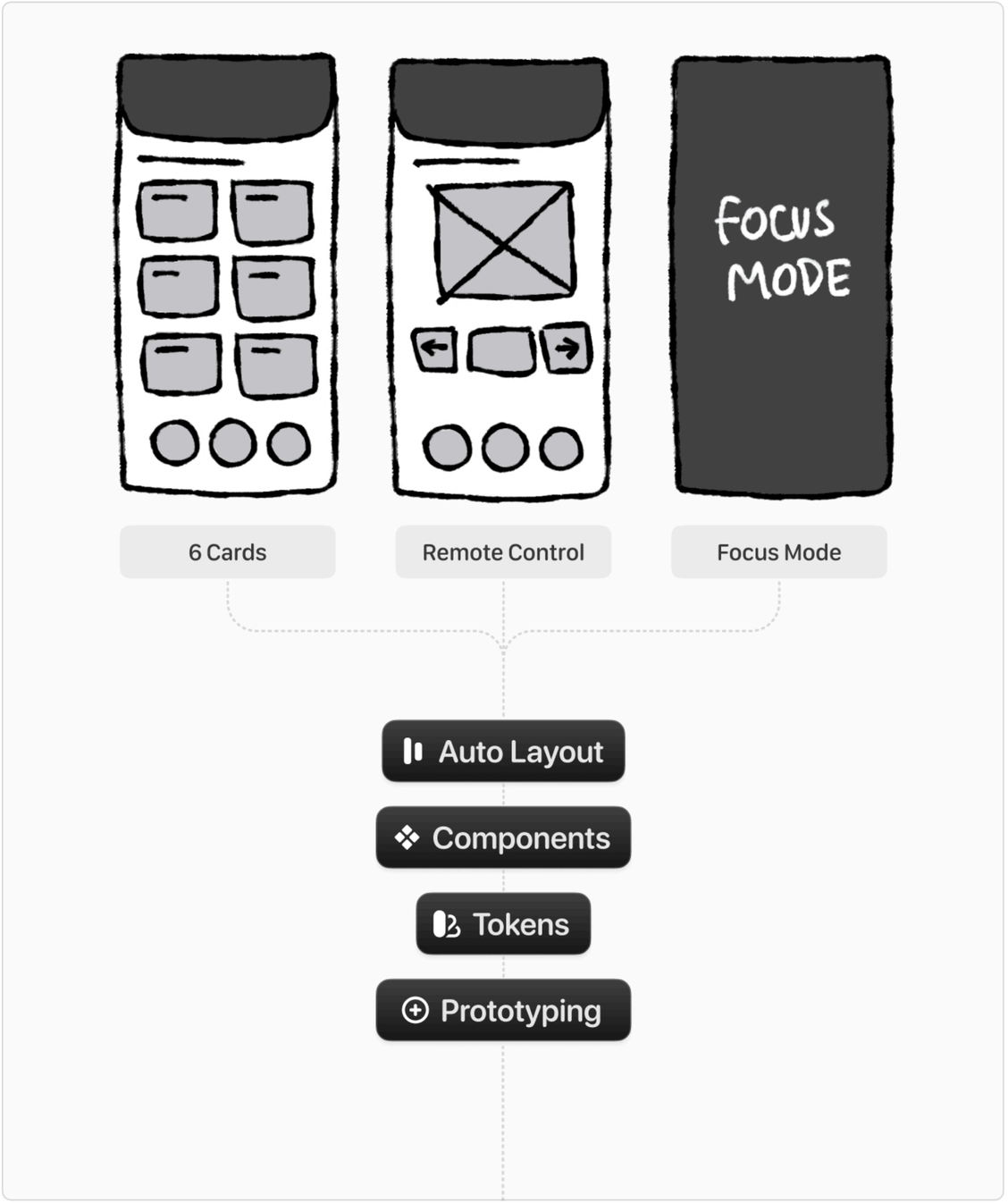
Rapid Architecture

To align the team without losing momentum, we drove rapid consensus through low-fidelity whiteboard wireframing, mapping the 4 pillars into a streamlined control interface with 6 modular UI cards.

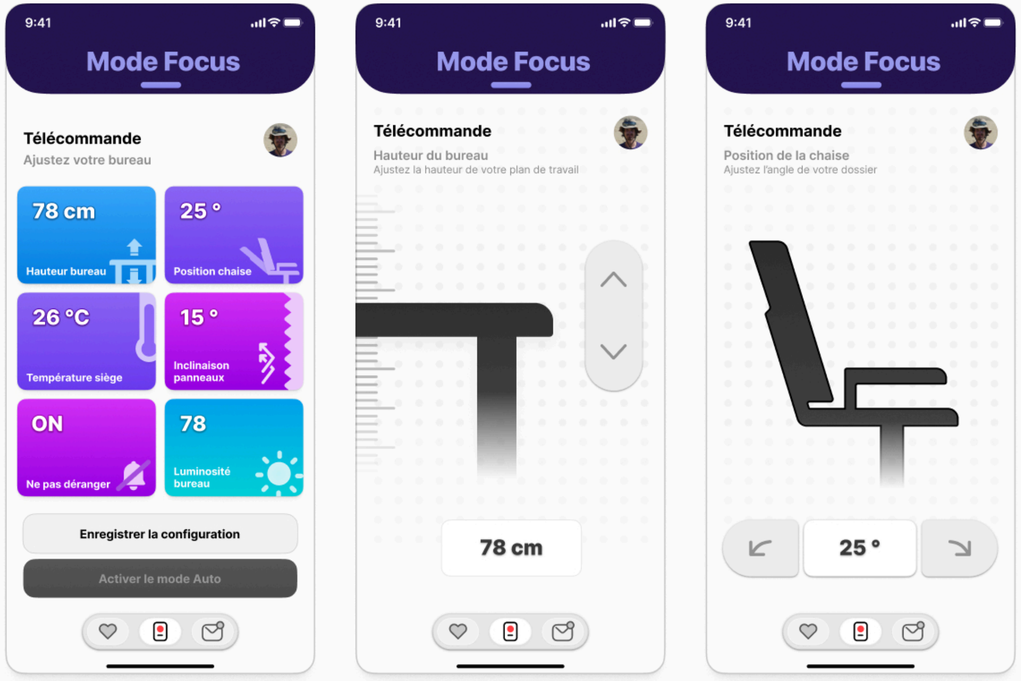
Prototyping

Stepping in as Visual Design Lead, I engineered the high-fidelity Figma prototype from scratch using Auto Layout rules, tokenized color palettes, and reusable component libraries.

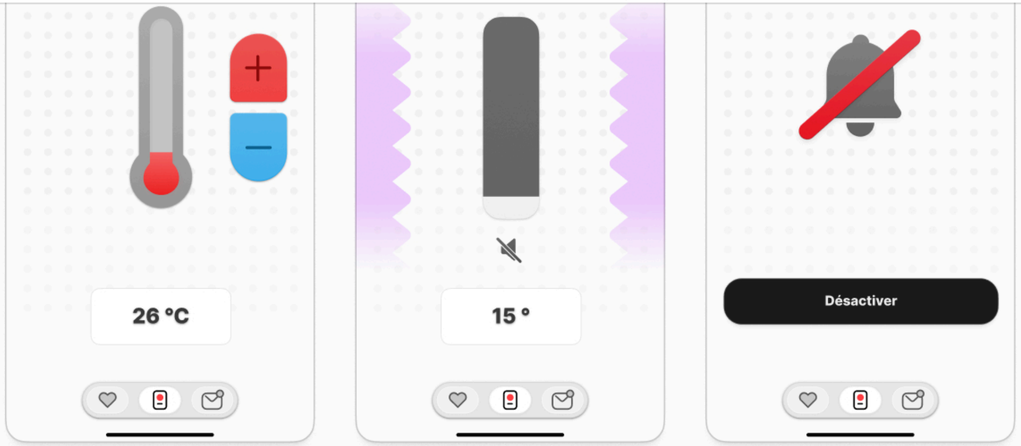
The flagship UX feature was a physical Focus Mode: users dock their smartphone onto the desk controller, transforming the phone into a tactile workspace remote while automatically filtering out non-essential digital notifications.



* UI microcopy rendered in native French deployment context



Smart Workspace Dashboard & Ergonomic Adjustments



Environmental Comfort & Acoustic Controls

Tactile Stepper UI

Ambient Lighting Control Widget

Contextual Focus Mode State

Testing Protocol

In the final phase, we evaluated an unbuilt 100L autonomous supermarket cart concept designed by a third team. I co-designed a guerrilla field usability test at a Leclerc supermarket in Pessac with 9 real shoppers.

To evaluate spatial navigation efficiently, we engineered a hybrid testing protocol pairing an interactive Figma mobile app with a physical cardboard store map and a cart token, allowing us to evaluate shopper mental models without requiring full-store physical walks.

9 shoppers

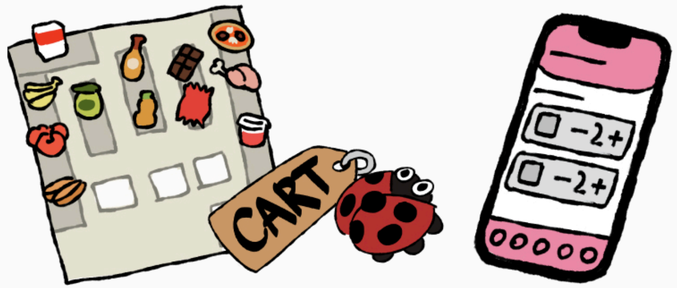
In-Situ Guerrilla Usability

Hybrid Test Setup

- Cardboard Map
- Mobile Prototype
- Cart Token

Captured Metrics (N=9)

- Task Success
- Click Count
- Path Efficiency
- Verbatims



Findings (Behavioral Realities)

Testing yielded a 100% completion rate for cart pairing but exposed foundational flaws in the product's premise. Five out of nine shoppers bypassed the digital checkout flow. Our protocol revealed a physical reflex trade-off: having a physical cart token in hand caused users' ingrained habit of pushing a cart to a cashier to override the digital UI prompt.

More critically, participants overwhelmingly stated a preference for paper grocery lists over the digital application, noting that the app caused them to "lose the initiative to think". Shoppers saw zero personal utility in the software, proving the digital feature set was a complex solution to an unvalidated problem.

5/9 Bypassed Digital Checkout

Ingrained Physical Reflex Override

Loss of Cognitive Initiative

Overwhelming Paper List Preference

Zero Perceived Utility

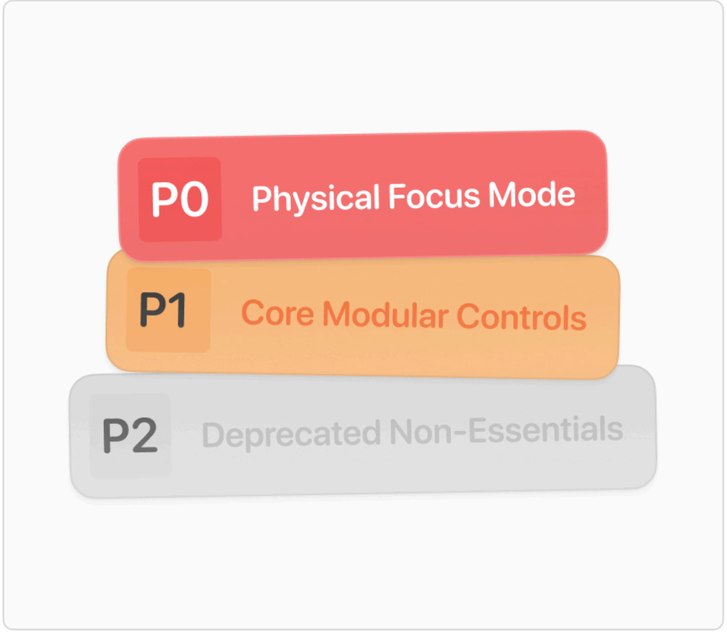
Complex Solution to Unvalidated Need

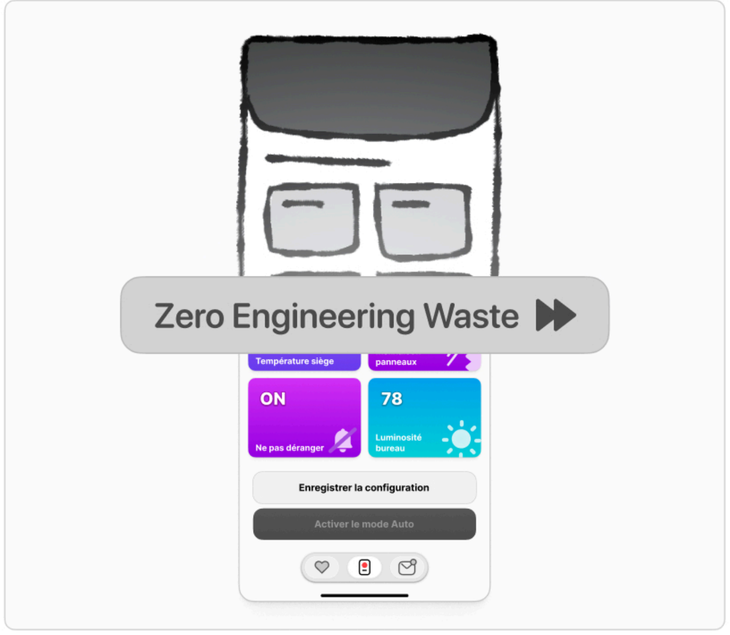
Impact and Reflections

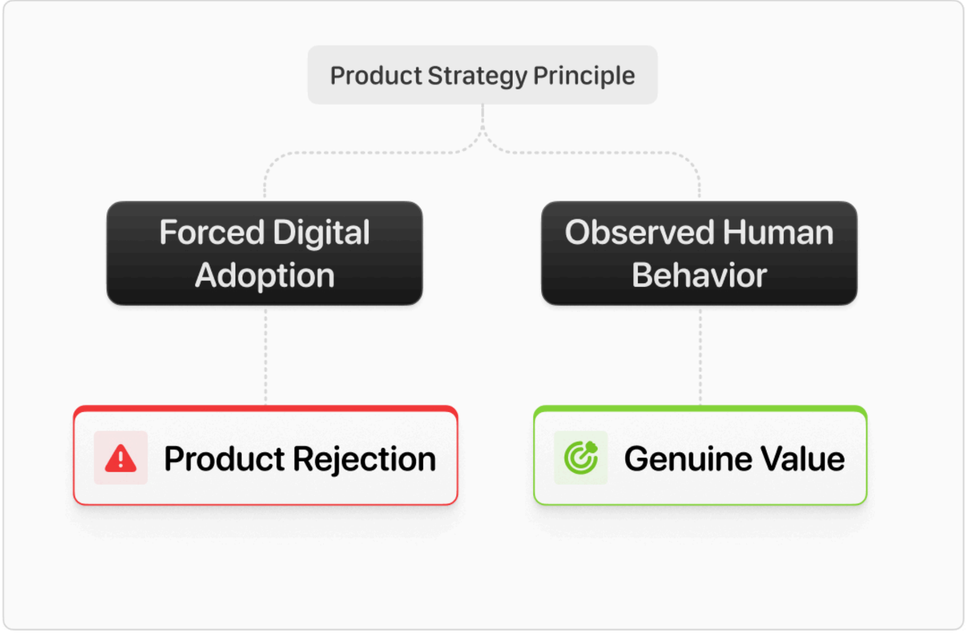
This design relay demonstrated that effective UX leadership requires ruthless prioritization and the courage to challenge flawed assumptions.

Navigating specification ambiguity in Phase 2 proved that structured low-fidelity wireframing prevents engineering waste and accelerates scalable Figma design systems.

Ultimately, Phase 3 reinforced that high-fidelity UI cannot salvage a product that fails to address a genuine human need: true product strategy prioritizes observing real human behavior over forcing unnecessary digital adoption.







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